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17	Design the following single table using HTML A B C D F E G H J K L M	2082/02/27	
18	Design the following table using nested tables in HTML. Logo (image) ABC Company: Home(link) About Us(link) Contact Us(link) Content 1 News 1 Content 2 News 2 Copyright Facebook, Twitter(link) Powered By	2082/02/27	
19	Design an html form to demonstrate the 10 different types of generated using INPUT tag	2082/02/27	
20	Design an html form to submit a job application form. Demonstrate use of drop-down list, text areas and file upload control.	2082/02/27	

CSS

S.N	LAB Questions	Date	Signature
1	Illustrate the use of background properties in CSS (for image) with four paragraph and style using CSS.	2082/02/27	
2	Design a web page with six images and style them using external CSS	2082/02/27	
3	Make a nested list using internal CSS	2082/02/27	
4	Make a web page having 3 rows and 2 columns with all the hyperlinks and style it using external CSS	2082/02/27	
5	Design an html form to demonstrate the 10 different types of controls generated using INPUT tag. Style the form as: - o Set the background color of textbox yellow when it is focused. - o Change the text color blue to textbox. - o Increase the height of combo options when it is checked. - o Set the border color red if the input is invalid in textbox textarea. - o Set green border to valid input. - o Design the form using fieldset	2082/02/27	
6	Design the given table and style as: Class A Class B Class C Class D - o Set the border of any style and color. - o Set width of all Classes 600 pixel and height 550 pixel. - o Set different background color for all classes. - o Every text color should be different. - o Display Class C and Class D in uppercase and remaining in lower case.	2082/02/27	
7	Design a horizontal menu of any five options and style as: - o Set border for all links. - o Set different background color and text color. - o Change the font color and increase the font size when mouse over it. - o Remove the underline from link. - o Change all the default behaviour of links.	2082/02/27	
8	Illustrate the concept of pseudo elements with example. (::after, ::first-letter, ::selection, ::firstline) and Develop a web page with a paragraph and add some contents before the paragraph with red color and yellow background using CSS.	2082/02/27	

JAVASCRIPT AND XML

S.N	Lab Questions	Date	Signature
1	Illustrate with examples the concept of alert, confirm and popup boxes.	2082/02/28	
2	WAP in javascript to demonstrate the objects, its properties and constructors with examples	2082/02/28	
3	How array can be created in javascript? Create 2-D array and display the array in HTML table.	2082/02/28	
4	Explain all the following built in objects with examples of each: o Array Methods o Date Methods o String Methods o Math Methods	2082/02/28	
5	WAP that contains a text field for name, age, phone number, email, dropdown for colleges, radio for gender and a checkboxes for i accept all the terms. Perform the client side validations as: o All fields are required o Name must contains alphabets and white spaces and of length greater than 6 characters long o Age must be integer and must be between and 50 o Phone number must be exactly 10 o digits long and must start with 98 or 97 o Email must be valid	2082/02/28	
6	WAP to show the concept of Try/Catch	2082/02/28	
7	Write jQuery to show the concept of on click, on mouse out, onkeyup, on form submit events	2082/02/28	
8	Illustrate how jQuery effects(hide/show, fade, stop can be done in any image)	2082/02/28	
9	How animation is done using jQuery? Illustrate with example	2082/02/28	
10	How json file is created? How can we parse the JSON file using JavaScript?	2082/02/28	
11	How AJAX is used to get the data from the server to the client side? Illustrate with example .	2082/02/29	
12	Create a simple XML element	2082/02/29	
13	Create an XML document that conforms to the following DTD	2082/02/29	

14	Create a XML document with simple and complex type	2082/02/29	
15	How XSLT is used to transform the XML document? Illustrate with example	2082/02/29	

PHP

S.N	Lab Questions	Date	Signature
1	Declare variables of different data types (integer, float, string) and demonstrate the usage of constants in PHP .	2082/02/30	
2	Create a PHP program that uses if-else statements to check if a number is positive, negative, or zero. Utilize different operators for comparison.	2082/02/30	
3	Develop a function that calculates the average of an array of numbers. Test the function with various arrays	2082/02/30	
4	Define a PHP class representing a "Book" with properties like title and author. Create objects of this class and display their attributes.	2082/02/30	
5	Design a simple HTML form with input fields. Create a PHP script that retrieves and displays the form data when submitted.	2082/02/30	
6	Enhance the previous form with PHP validation. Ensure that required fields are filled, and validate email addresses and make sure the phone number starts with 98 or 97 using regex	2082/02/31	
7	Implement PHP code that responds to a button click event on a form. Display a message when the button is clicked	2082/02/31	
8	Create a PHP script that sets a cookie with user preferences.Retrieve and display this information on subsequent visits	2082/02/31	
9	Build a form with various elements (textboxes, checkboxes, radio buttons). Use PHP to access and display the selected values .	2082/02/31	
10	Develop a PHP script that reads content from a text file, modifies it, and then writes the updated content back to the file	2082/02/31	

HTML

1. Write a complete source code to demonstrate the use of html, head, title, body tag and some content

Source code:

```
<!DOCTYPE html>

<html lang="en">

  <head>

    <meta charset="UTF-8" />

    <meta name="viewport" content="width=device-width, initial-scale=1.0" />

    <title>Introduction of HTML</title>

  </head>

  <body>

    <h3>

      Write a complete source code to demonstrate the use of html, head, title, body tag and some content

    </h3>

    <p>My name is Arjun Mijar.</p>

    <div>

      <p>Multiple  spaces  are collapsed.</p>

      <p>Leading and trailing

        spaces are preserved.</p>

      <p>Line breaks are also preserved.</p>

    </div>

  </body>

</html>
```

Output:



Write a complete source code to demonstrate the use of html, head, title, body tag and some content

My name is Arjun Mijar.

Multiple spaces are collapsed.

Leading and trailing spaces are preserved.

Line breaks are also preserved.

2. Create two paragraphs. Paragraph 1 containing the heading as Management Colleges in Nepal and Paragraph 2 containing the heading as Engineering Colleges in Nepal. Heading of Paragraph 2 must be smaller than the heading of Paragraph 1. Also show the use of b, i, sup, sub, small, strong, abbr, acronym, blockquote with cite attribute, city, q tag inside the two paragraphs.

Source code:

```
<!DOCTYPE html>

<html>

<head>

  <title>Colleges in Nepal</title>

</head>

<body>


  <!-- Paragraph 1: Management Colleges -->

  <h2>Management Colleges in Nepal</h2>

  <p>

    <strong>Kathmandu College of Management</strong> is a reputed <b>management</b> college located in <city>Lalitpur</city>. It is known for producing <i>business leaders</i> in Nepal.

    Students often refer to <abbr title="Bachelor of Business Administration">BBA</abbr> and <acronym title="Master of Business Administration">MBA</acronym> programs.

    The college was established in <small>1997</small>, and is still one of the <strong>top choices</strong> for management education.<br>

    A quote from a student: <q>Studying at KCM was a life-changing experience.</q><br>

    <blockquote cite="https://kcm.edu.np">

      KCM emphasizes leadership and innovation in business education.

    </blockquote>

  </p>


  <!-- Paragraph 2: Engineering Colleges -->

  <h3>Engineering Colleges in Nepal</h3>

  <p>

    One of the top engineering colleges is <strong>Pulchowk Campus</strong> located in <city>Lalitpur</city>.
```

It offers degrees in **Civil**, *Computer*, and **Electrical** engineering.

Most students pursue BE degrees.

Pulchowk also offers MSc programs in advanced technologies.

The college was founded in 1972.

A scientific expression used is: H_2O and $x^2 + y^2 = r^2$.

A quote from a professor: "Engineering is the backbone of national development."

<https://pcampus.edu.np>

Pulchowk Campus focuses on technical excellence and innovation.

Output:

← → ↺

File

D:/Arjun%20Mijar(109)%20Lab%20Reports/Web%20Technology/Q2.html

☆ 📄 📁 👤 ⋮

Management Colleges in Nepal

Kathmandu College of Management is a reputed **management** college located in Lalitpur. It is known for producing *business leaders* in Nepal. Students often refer to **BBA** and **MBA** programs. The college was established in 1997, and is still one of the **top choices** for management education.

A quote from a student: "Studying at KCM was a life-changing experience."

KCM emphasizes leadership and innovation in business education.

Engineering Colleges in Nepal

One of the top engineering colleges is **Pulchowk Campus** located in Lalitpur. It offers degrees in **Civil**, *Computer*, and **Electrical** engineering. Most students pursue **BE** degrees.

Pulchowk also offers MSc programs in advanced technologies.

The college was founded in 1972.

A scientific expression used is: H_2O and $x^2 + y^2 = r^2$.

A quote from a professor: "Engineering is the backbone of national development."

Pulchowk Campus focuses on technical excellence and innovation.

3. Write the HTML code to display the following output:

**Hi,
My
Name
Is
John**

And,

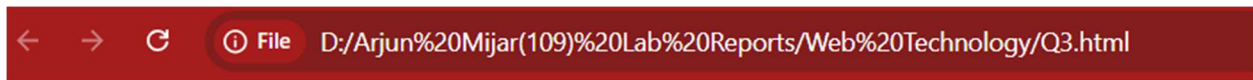
- a. code**
- b. kbd**
- c. var**
- d. samp**

Source code:

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8" />
  <meta name="viewport" content="width=device-width, initial-scale=1.0" />
  <title>Demonstrate pre and list</title>
</head>
<body>
  <div>
    <pre>
      Hi,
      My
      Name
      Is
    </pre>
  </div>
  <div>
    <ol type="a">
      <li>code</li>
      <li>kdb</li>
```

```
<li>var</li>
<li>samp</li>
</ol>
</div>
</body>
</html>
```

Output:



Hi,
My
Name
Is

- a. code
- b. kdb
- c. var
- d. samp

4. Write a HTML code to demonstrate the concept of:

- o Unordered list**
- o ordered list**
- o Nested unordered list**
- o Nested ordered list**
- o Definition list**

Source code:

```
<!DOCTYPE html>

<html>

<head>

  <title>HTML Lists Example</title>

</head>

<body>


  <!-- Unordered List -->

  <h2>Unordered List</h2>

  <ul>

    <li>Apple</li>

    <li>Banana</li>

    <li>Mango</li>

  </ul>


  <!-- Ordered List -->

  <h2>Ordered List</h2>

  <ol>

    <li>First Step</li>

    <li>Second Step</li>

    <li>Third Step</li>

  </ol>


  <!-- Nested Unordered List -->
```

<h2>Nested Unordered List</h2>

Fruits

Apple

Orange

Vegetables

Carrot

Potato

<!-- Nested Ordered List -->

<h2>Nested Ordered List</h2>

HTML

Head

Body

CSS

Selectors

Properties

<!-- Definition List -->

<h2>Definition List</h2>

<dl>

<dt>HTML</dt>

<dd>HyperText Markup Language</dd>

<dt>CSS</dt>

<dd>Cascading Style Sheets</dd>

<dt>JS</dt>

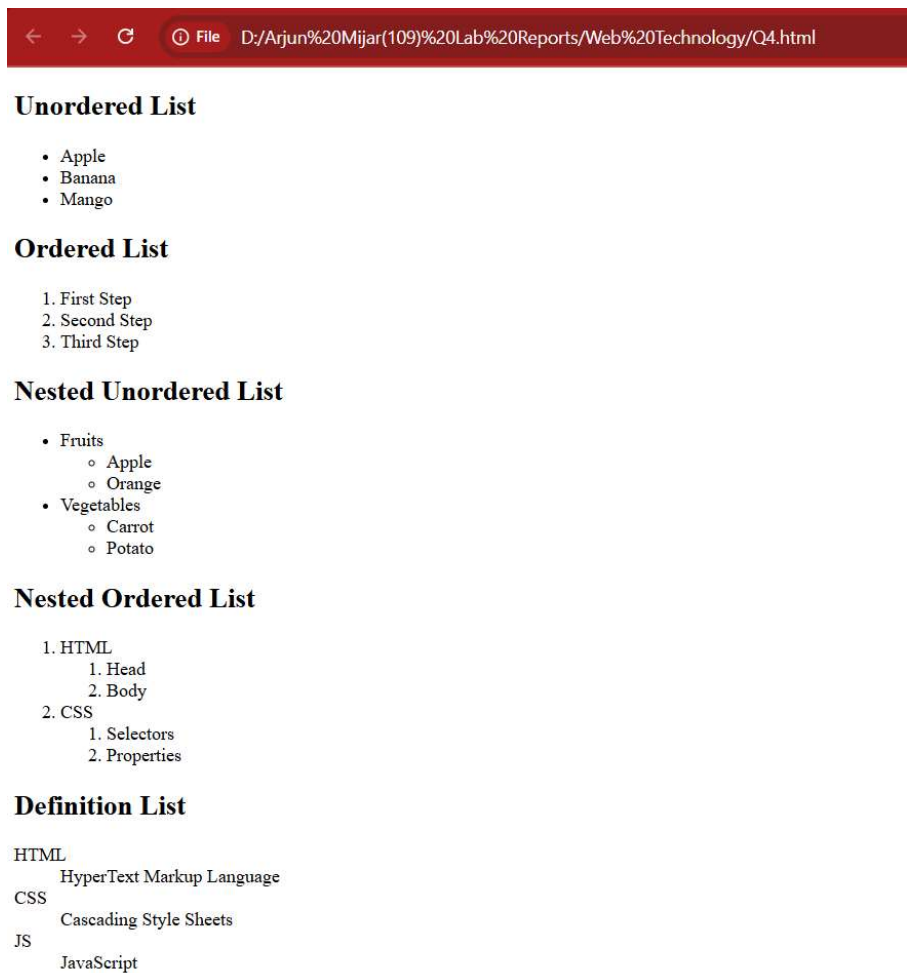
<dd>JavaScript</dd>

</dl>

</body>

</html>

Output:



5. Write the HTML code to demonstrate image as list

Source code:

```
<!DOCTYPE html>

<html>

  <head>
    <title>Image as List Items</title>
  </head>

  <body>
    <h2>Image List (Unordered):</h2>
    <ul>
      <li>
        
      </li>
      <li>
        
      </li>
    </ul>
    <h2>Image List (Ordered):</h2>
    <ol>
      <li>
        
      </li>
      <li>
        
      </li>
    </ol>
```

```
</body>  
</html>
```

Output:



Image List (Unordered):

- 
- 

Image List (Ordered):

1. 
2. 

6. Write the HTML code to demonstrate the use of URL along with Absolute URL, Relative URL and also create a link

Source code:

```
<!DOCTYPE html>

<html lang="en">

  <head>

    <meta charset="UTF-8" />

    <meta name="viewport" content="width=device-width, initial-scale=1.0" />

    <title>Document</title>

  </head>

  <body>

    <h2>Absolute URL Example:</h2>

    <p>

      <a href="https://www.example.com" target="_blank">Visit Example.com</a>

    </p>

    <h2>Relative URL Example:</h2>

    <p>

      <a href="8_list.html">Go to a Relative Page</a>

    </p>

    <h2>Relative URL within Subdirectory:</h2>

    <p>

      <a href="subdirectory/another-page.html" >Go to Another Page in Subdirectory</a>

    </p>

    <nav>

      Link

      <ul type="none">

        <li><a href=" ../home.html">Home</a></li>

        <li><a href=" ../about/home.html">About</a></li>

        <li><a href="/gallery.html">Gallery</a></li>

      </ul>
```

```
</nav>  
</body>  
</html>
```

Output:



Absolute URL Example:

[Visit Example.com](http://example.com)

Relative URL Example:

[Go to a Relative Page](#)

Relative URL within Subdirectory:

[Go to Another Page in Subdirectory](#)

Link

[Home](#)
[About](#)
[Gallery](#)

7. Create an image gallery of 9 images with 3 images in a row each having 200 width and height 250.

Source code:

```
<!DOCTYPE html>

<html lang="en">

  <head>

    <meta charset="UTF-8" />

    <meta name="viewport" content="width=device-width, initial-scale=1.0" />

    <title>Image</title>

  </head>

  <body>

    <h2>Image Gallery</h2>

    <div>

    </div>

    <div>

    </div>

    <div>

    </div>

  </body>

</html>
```

Output:

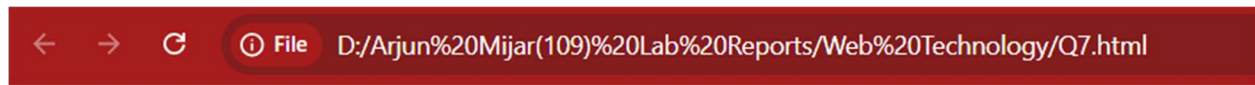
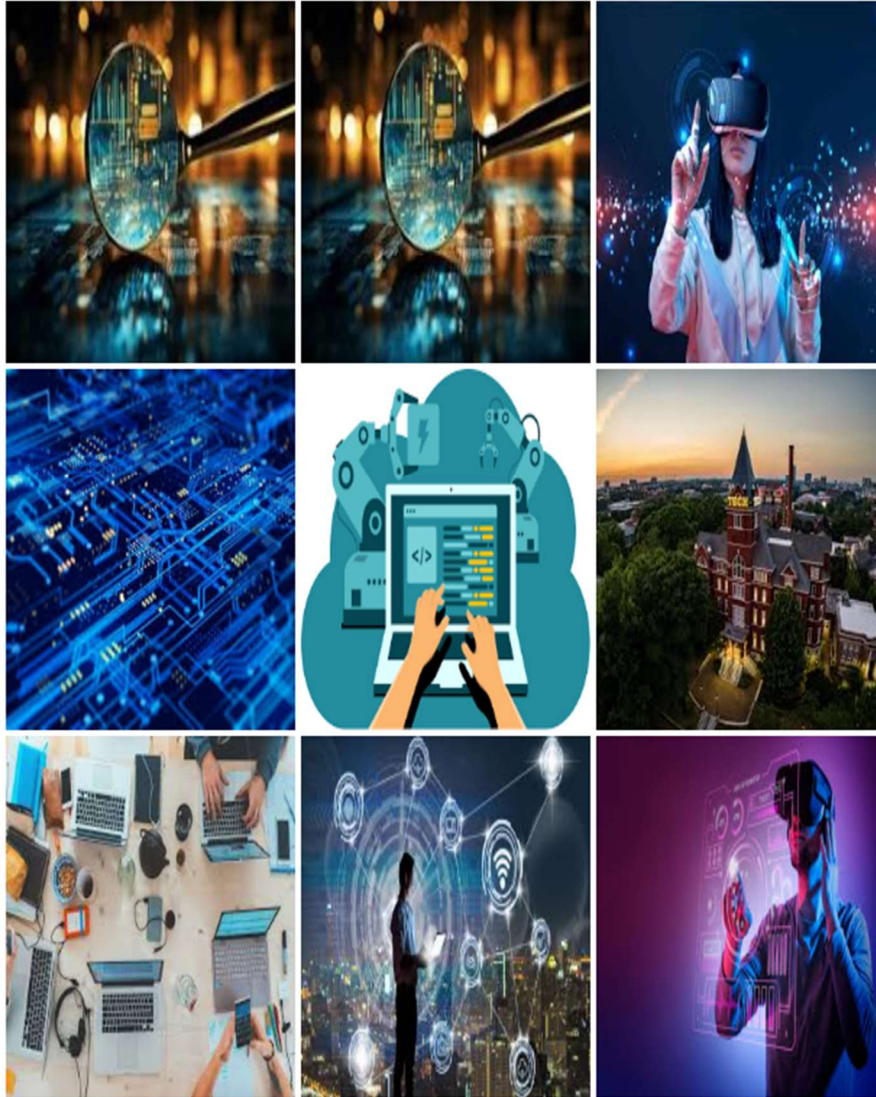


Image Gallery



8. Write a HTML code to map image as Client side and ServerSide and demonstrate how audio, video can be added to the web pages along with all the attributes

Source code:

```
<!DOCTYPE html>

<html lang="en">

  <head>

    <meta charset="UTF-8" />

    <meta name="viewport" content="width=device-width, initial-scale=1.0" />

    <title>image with clint and server script</title>

  </head>

  <body>

    <h2>Click on the image to see client-side and server-side actions</h2>

    <map name="workmap">

      <area shape="rect" coords="34,44,270,350" alt="Computer" href="Q6.html" />

      <area shape="rect" coords="290,172,333,250" alt="Phone" href="Q5.html" />

      <area shape="circle" coords="337,300,44" alt="Coffee" href="Q7.html" />

    </map>

    <div>

      <!-- Audio Example -->

      <div>

        <p>Song mp3</p>

        <audio controls preload="auto" loop>

          <source src="media/song.mp3" type="audio/mp3" />

          Your browser does not support the audio element.

        </audio>

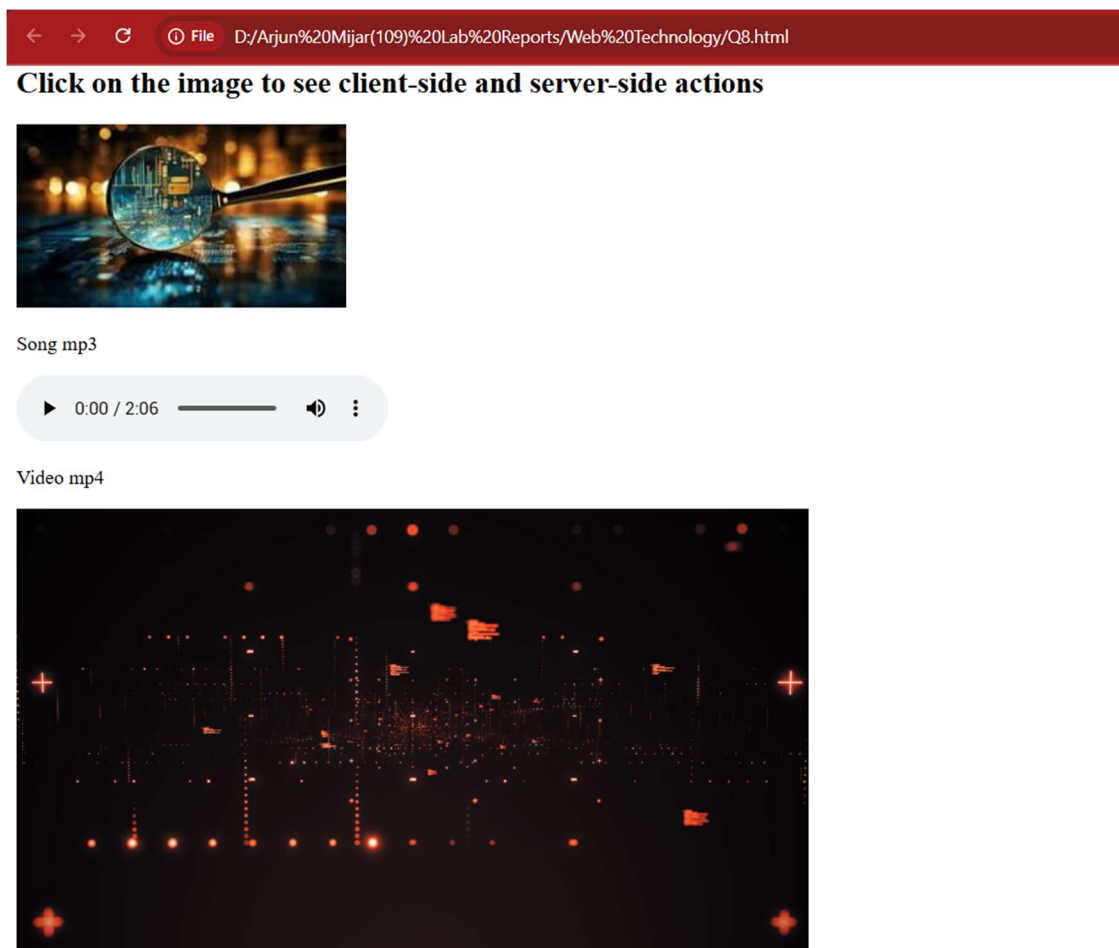
      </div>

      <!-- Video Example -->

      <div>
```

```
<p>Video mp4</p>
<video width="640" height="360" controls autoplay muted>
  <p>Video mp4</p>
  <source src="media/video.mp4" type="video/mp4" />
  Your browser does not support the video element.
</video>
</div>
</div>
</body>
</html>
```

Output:



9. Write the HTML code to print the following output:

SN	Name		College	Address	Email	Phone
	First Name	Last Name				

Fill atleast 5 details.

Source code:

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>table</title>
</head>
<body>
  <table border="1" width="100%" cellpadding="10">
    <tr>
      <th rowspan="2">SN</th>
      <th colspan="2">Name</th>
      <th rowspan="2">College</th>
      <th rowspan="2">Address</th>
      <th rowspan="2">Email</th>
      <th rowspan="2">Phone</th>
    </tr>
    <tr>
      <th>First Name</th>
      <th>Last Name</th>
    </tr>
    <tr>
      <td>1</td>
      <td>Arjun</td>
      <td>Bijaya</td>
```

<td>Devinagar</td>
<td>Kathmandu</td>
<td>arjun@gmail.com</td>
<td>9841984198</td>
</tr>
<td>2</td>
<td>Aayush</td>
<td>Bibek</td>
<td>Maitidevi</td>
<td>Kathmandu</td>
<td>bibek@gmail.com</td>
<td>9841984598</td>
</tr>
<td>3</td>
<td>Bi pana</td>
<td>Kushal</td>
<td>Pulchowk</td>
<td>Lalitpur</td>
<td>kushal@gmail.com</td>
<td>9841984198</td>
</tr>
<td>4</td>
<td>Niva</td>
<td>Maharjan</td>
<td>khusibu</td>
<td>Kathmandu</td>
<td>nivv@gmail.com</td>

```

<td>9841984198</td>

</tr>

<tr>

<td>5</td>

<td>Kabir</td>

<td>Saheb</td>

<td>satlok</td>

<td>Kathmandu</td>

<td>kabir@gmail.com</td>

<td>9841954198</td>

</tr>

</table>

</body>

</html>

```

Output:

SN	Name		College	Address	Email	Phone
	First Name	Last Name				
1	Arjun	Bijaya	Devinagar	Kathmandu	arjun@gmail.com	9841984198
2	Aayush	Bibek	Maitidevi	Kathmandu	bibek@gmail.com	9841984598
3	Bipana	Kushal	Pulchowk	Lalitpur	kushal@gmail.com	9841984198
4	Niva	Maharjan	khusibu	Kathmandu	nivv@gmail.com	9841984198
5	Kabir	Saheb	satlok	Kathmandu	kabir@gmail.com	9841954198

10. Create a simple html page to demonstrate the basic structural tags of HTML

Source code:

```
<!DOCTYPE html>

<html>

<head>

  <title>Basic HTML Structure</title>

</head>

<body>

  <!-- Header Section -->

  <header>

    <h1>Welcome to My Web Page</h1>

    <p>This is a demonstration of HTML structural tags.</p>

  </header>

  <!-- Navigation Section -->

  <nav>

    <ul>

      <li><a href="#">Home</a></li>

      <li><a href="#">About</a></li>

      <li><a href="#">Contact</a></li>

    </ul>

  </nav>

  <!-- Main Content Section -->

  <main>

    <section>

      <h2>About HTML</h2>

      <p>HTML stands for HyperText Markup Language. It is used to create web pages.</p>

    </section>

    <article>

      <h2>Latest Update</h2>

      <p>This article talks about the new features in HTML5.</p>
```

```
</article>

</main>

<!-- Aside Section -->

<aside>

    <h3>Did You Know?</h3>

    <p>HTML5 introduced new semantic elements like <header>, <footer>, <article>,
    <section>, and <aside>.</p>

</aside>

<!-- Footer Section -->

<footer>

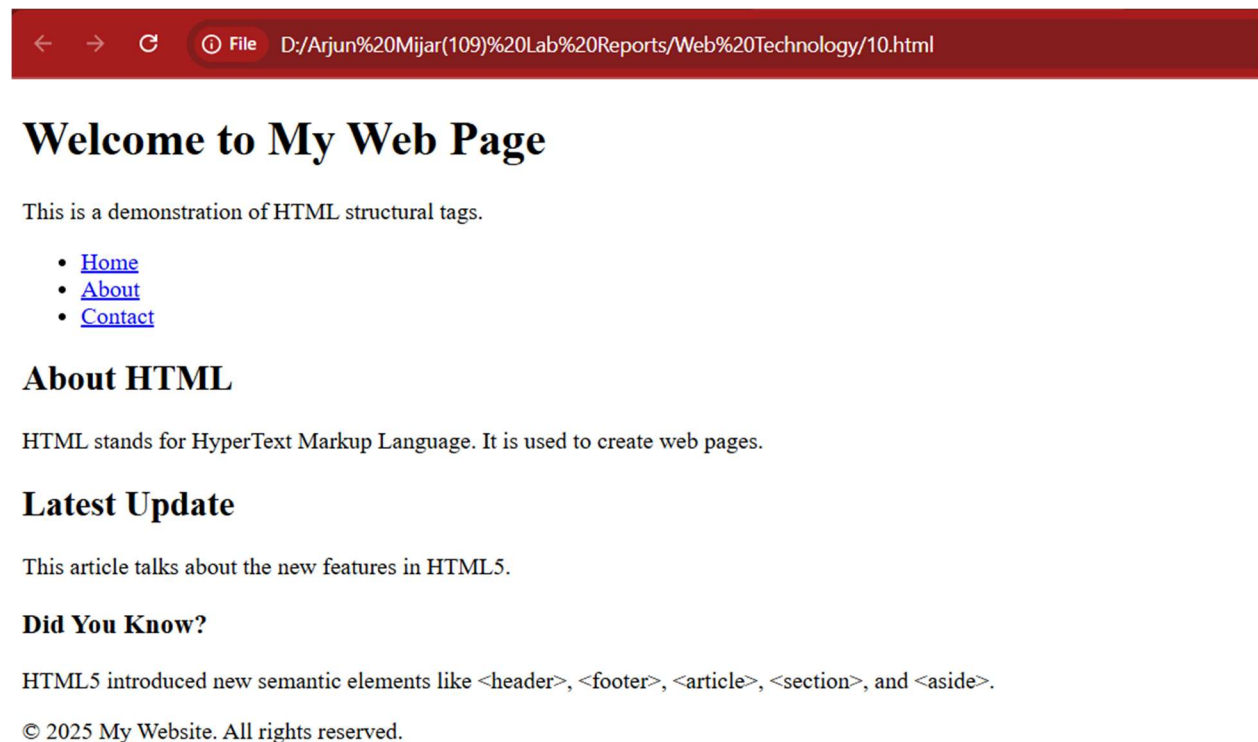
    <p>&copy; 2025 My Website. All rights reserved.</p>

</footer>

</body>

</html>
```

Output:



11. Prepare an html page to display the syntax and usage of any 10 text formatting tags.

Source Code:

```
<!DOCTYPE html>

<html>

<head>

  <title>Text Formatting Tags in HTML</title>

</head>

<body>

  <h2>Text Formatting Tags with Syntax and Usage</h2>

  <p><b>&lt;b>Bold Text&lt;/b> →</b> This text is bold.</p>

  <p><strong>&lt;strong>Strong Text&lt;/strong> →</strong> This text is important.</p>

  <p><i>&lt;i>Italic Text&lt;/i> →</i> This text is italicized.</p>

  <p><em>&lt;em>Emphasized Text&lt;/em> →</em> This text is emphasized.</p>

  <p><u>&lt;u>Underlined Text&lt;/u> →</u> This text is underlined.</p>

  <p><mark>&lt;mark>Marked Text&lt;/mark> →</mark> This text is highlighted.</p>

  <p>This is normal<small> &lt;small>small text&lt;/small></small>.</p>

  <p>H<sub>2</sub>O → Uses &lt;sub> for subscript.</p>

  <p>X<sup>2</sup> → Uses &lt;sup> for superscript.</p>

  <p><del>&lt;del>Deleted Text&lt;/del> and <ins>&lt;ins>Inserted
Text&lt;/ins> are useful for revisions.</p>

</body>

</html>
```

Output:

← → ↻ ⓘ File D:/Arjun%20Mijar(109)%20Lab%20Reports/Web%20Technology/Q11.html

Text Formatting Tags with Syntax and Usage

Bold Text → This text is bold.

Strong Text → This text is important.

<i>Italic Text</i> → This text is italicized.

Emphasized Text → This text is emphasized.

<u>Underlined Text</u> → This text is underlined.

<mark>Marked Text</mark> → This text is highlighted.

This is normal <small>small text</small>.

H₂O → Uses `<sub>` for subscript.

X² → Uses `<sup>` for superscript.

~~Deleted Text~~ and <ins>Inserted Text</ins> are useful for revisions.

12. Create an html page to write brief description about your college using header tags, paragraph tag and text formatting tags.

Source code:

```
<!DOCTYPE html>

<html>

<head>

  <title>About ASCOL</title>

</head>

<body>


  <!-- Main Heading -->

  <h1>Amrit Science College (ASCOL)</h1>


  <!-- Subheading -->

  <h2>Introduction</h2>

  <p>

    <strong>Amrit Science College</strong>, commonly known as <b>ASCOL</b>, is one of the oldest
    and most prestigious science colleges in Nepal. It is affiliated with <i>Tribhuvan University</i> and is
    located in <u>Lainchaur, Kathmandu</u>.

  </p>


  <!-- Subheading -->

  <h2>Academic Excellence</h2>

  <p>

    ASCOL offers a variety of undergraduate and postgraduate programs in the field of science such as
    <em>BSc CSIT</em>, <em>BSc Microbiology</em>, and <em>MSc Chemistry</em>. It is known for
    producing academically strong and highly disciplined students.

  </p>


  <!-- Subheading -->

  <h2>Facilities and Environment</h2>

  <p>
```

The college provides well-equipped laboratories, a peaceful library, and a <mark>student-friendly learning environment</mark>. ASCOL encourages students to be innovative and research-oriented.

</p>

<!-- Subheading -->

<h2>Legacy and Reputation</h2>

<p>

Established in <small>1962</small>, ASCOL has a long legacy of excellence in science education. Many renowned scientists and professionals in Nepal have graduated from this college.

</p>

<!-- Subheading -->

<h2>Conclusion</h2>

<p>

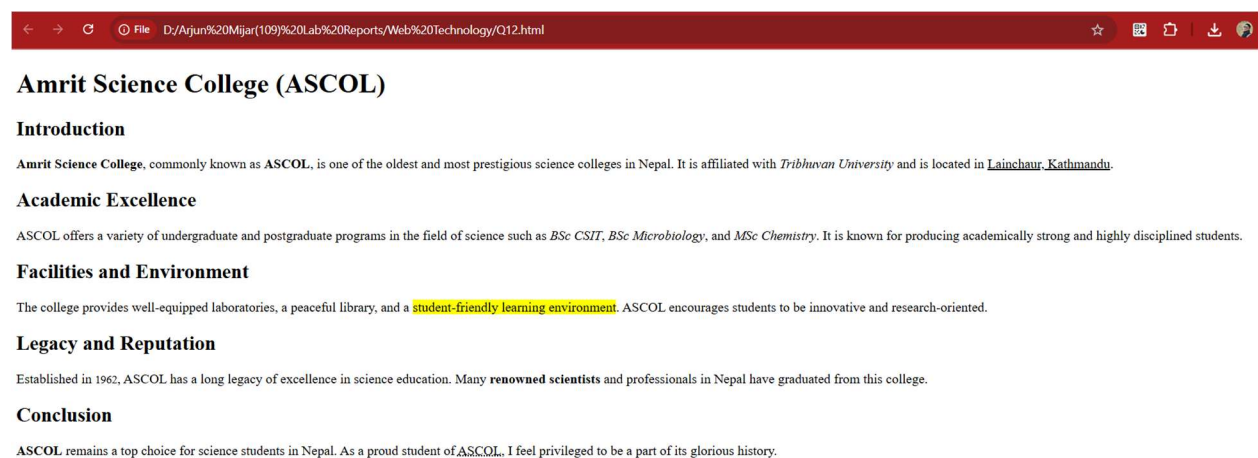
ASCOL remains a top choice for science students in Nepal. As a proud student of <abbr title="Amrit Science College">ASCOL</abbr>, I feel privileged to be a part of its glorious history.

</p>

</body>

</html>

Output:



13. Create an image gallery of any three popular places in Nepal along with their short description with the images aligned in left or right.

Source code:

```
<!DOCTYPE html>

<html>

<head>

  <title>Popular Places in Nepal</title>

</head>

<body>

  <h1>Popular Places in Nepal</h1>

  <!-- Place 1: Pokhara -->

  <h2>Pokhara</h2>

  <p>

    Pokhara is a beautiful city known for its lakes, mountains, and adventure tourism. It is the gateway to
the Annapurna trekking circuit and offers serene views of the Himalayas.

  </p>

  <br clear="all"><hr>

  <!-- Place 2: Lumbini -->

  <h2>Lumbini</h2>

  <p>

    Lumbini is the birthplace of Lord Buddha and a UNESCO World Heritage Site. It is an important
spiritual destination attracting pilgrims and tourists from around the world.

  </p>

  <br clear="all"><hr>
```

<!-- Place 3: Bhaktapur Durbar Square -->

<h2>Bhaktapur Durbar Square</h2>

<p>

Bhaktapur Durbar Square is an ancient royal palace courtyard rich in Newari architecture and culture. It showcases traditional art, temples, and historical monuments.

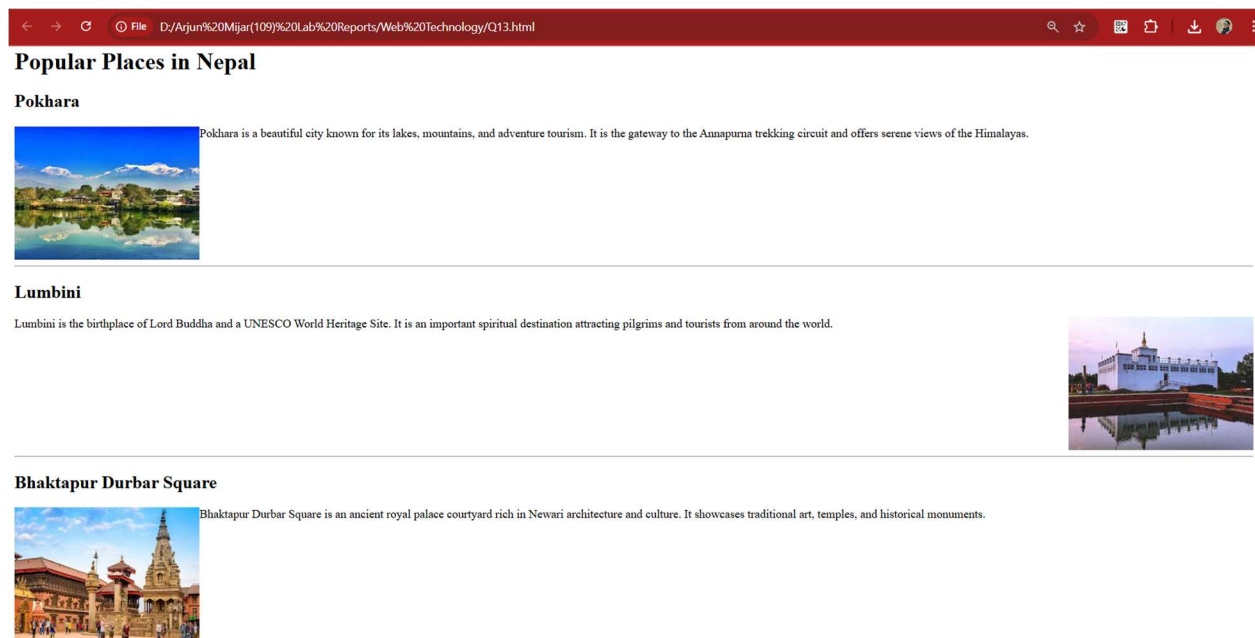
</p>

<br clear="all">

</body>

</html>

Output:



**14. Demonstrate the use of align, vspace and hspace attribute using minimum of six images.
(Make a gallery)**

Source code:

```
<!DOCTYPE html>
```

```
<html>
```

```
<head>
```

```
  <title>Image Gallery using align, vspace, hspace</title>
```

```
</head>
```

```
<body>
```

```
  <h2>Image Gallery Demonstrating align, vspace, hspace</h2>
```

```
  <!-- Image 1 -->
```

```
  
```

```
  
```

```
  <p>
```

These two images are aligned left and right with space around them using `<code>vspace</code>` and `<code>hspace</code>`. Observe the space above, below, and beside the images.

```
  </p>
```

```
  <br clear="all"><hr>
```

```
  <!-- Image 3 -->
```

```
  
```

```
  <!-- Image 4 -->
```

```
  
```

```
  <!-- Image 5 -->
```

```
  
```

```
  <p>
```

These three images show different vertical alignments: `<code>top</code>`, `<code>middle</code>`, and `<code>bottom</code>` with horizontal and vertical spacing.

```
  </p>
```

<!-- Image 6 -->

$\langle p \rangle$

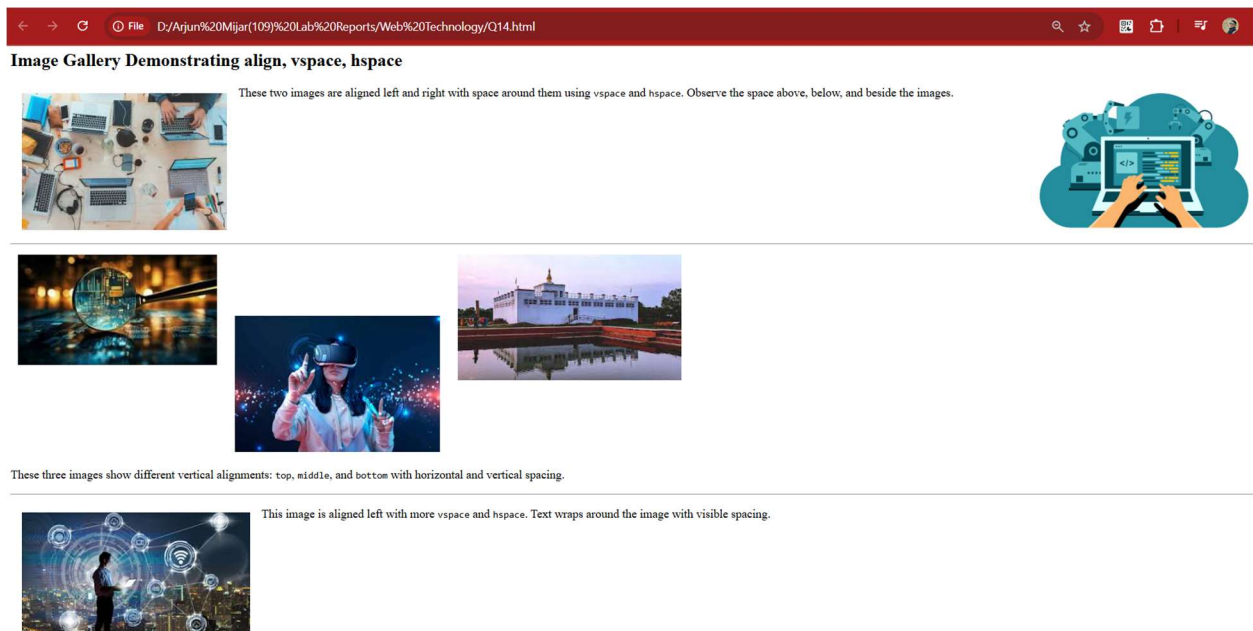
This image is aligned left with more `vspace` and `hspace`. Text wraps around the image with visible spacing.

</p>

```
<br clear="all">
```

</body>

Output:



15. Demonstrate the use of top to bottom and bottom to top link using anchor tags.

Source code:

```
<!DOCTYPE html>

<html>

<head>

  <title>Anchor Tag Navigation</title>

</head>

<body>

  <!-- Top Section -->

  <h1 id="top">Welcome to My Page</h1>

  <p>This page demonstrates top-to-bottom and bottom-to-top linking using anchor tags.</p>

  <!-- Link to Bottom -->

  <a href="#bottom">Go to Bottom of the Page</a>

  <hr>

  <!-- Dummy Content to Scroll -->

  <p>Lorem ipsum dolor sit amet, consectetur adipiscing elit. Sed et sapien sed mauris efficitur facilisis.</p>

  <p>Vestibulum ante ipsum primis in faucibus orci luctus et ultrices posuere cubilia Curae;</p>

  <p>Mauris ut tortor non risus consectetur tincidunt. Nulla facilisi.</p>

  <p>Sed mattis, metus nec fermentum gravida, magna ex tristique metus, at egestas lorem elit in justo.</p>

  <p>...</p>

  <p>...</p>

  <p>Scroll down for more content.</p>

  <!-- Bottom Section -->

  <h2 id="bottom">You Are Now at the Bottom!</h2>
```

<p>This is the bottom of the page.</p>

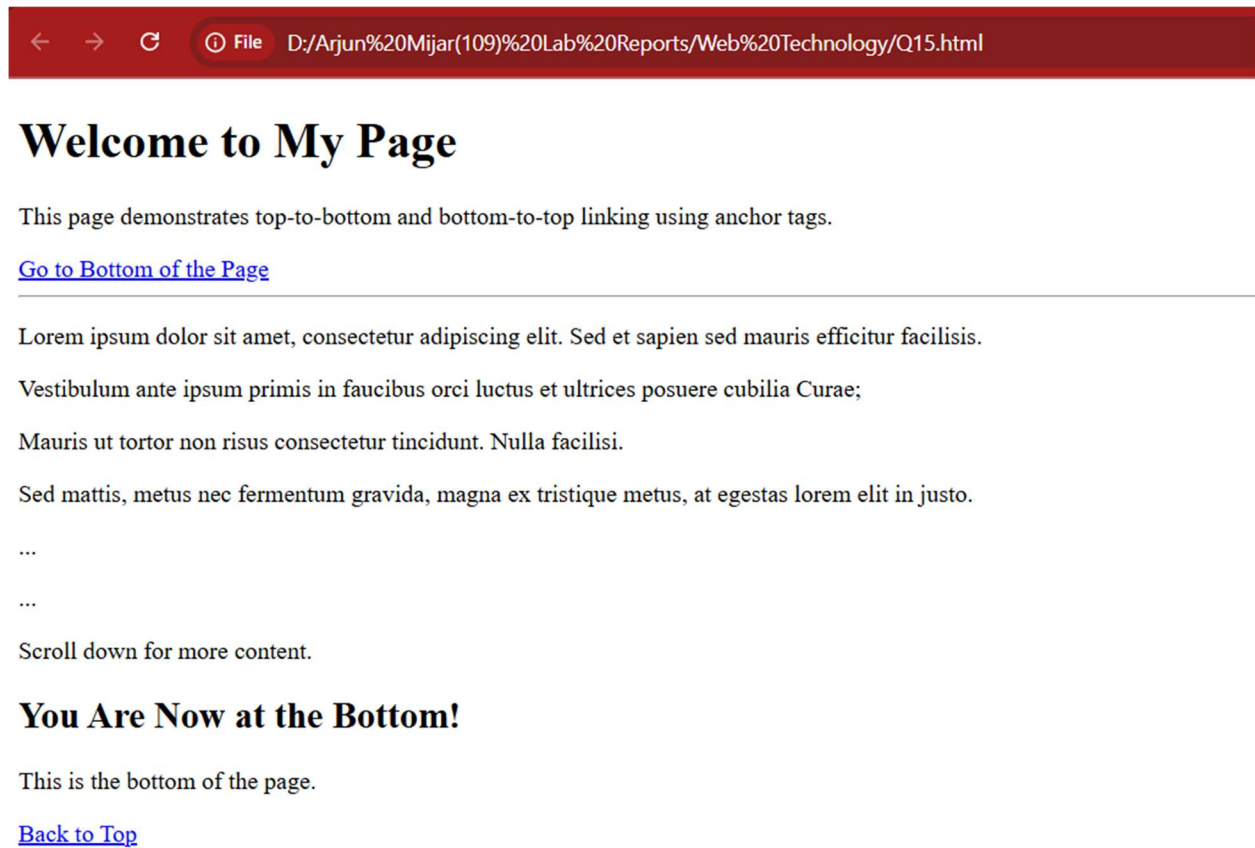
<!-- Link to Top -->

Back to Top

</body>

</html>

Output:



16. Create the following list using html.

1. BSCIT

2. MSCIT

a. Semester 1

▪ Subject 1

▪ Subject 2

▪ Subject 3

b. Semester 2

c. Semester 3

3. BScNT

And,

A. Movies

i. Nepali

ii. Hindi

iii. English

B. Music

C. Travel

D. Reading

E. Social Networking

F. Chatting

Source code:

```
<!DOCTYPE html>
```

```
<html lang="en">
```

```
<head>
```

```
  <meta charset="UTF-8">
```

```
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
```

```
  <title>List</title>
```

```
</head>
```

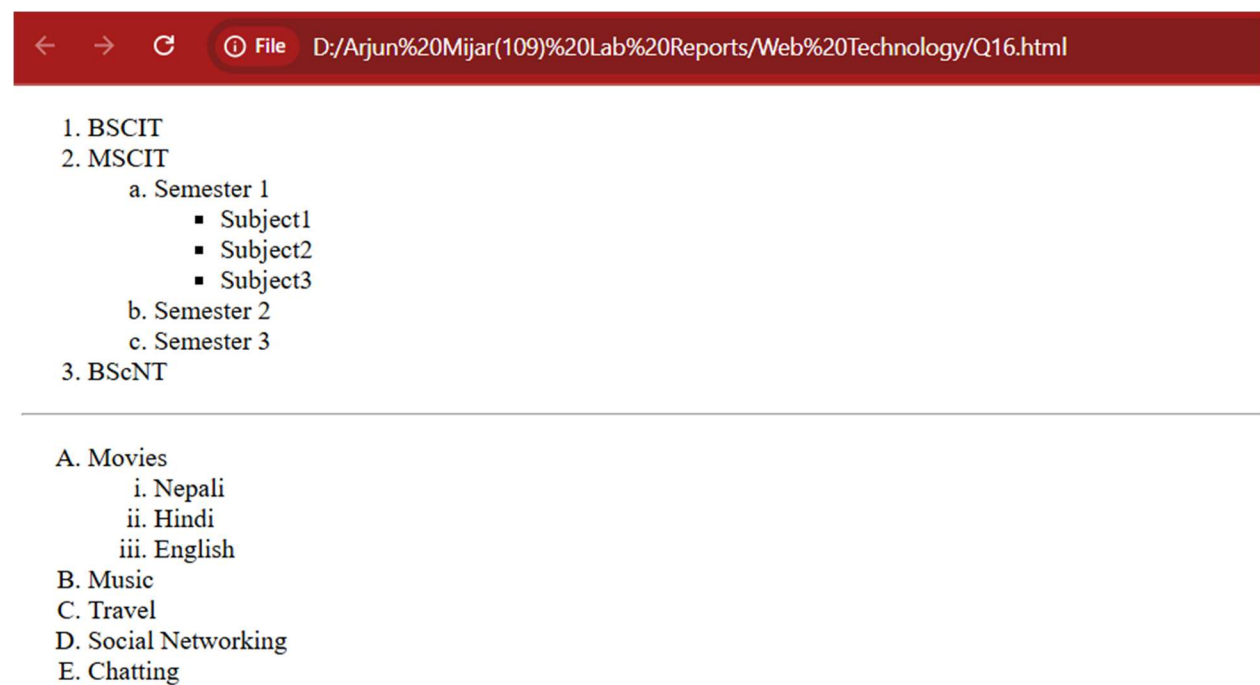
```
<body>
```

```
<div id="unorder-list">
  <ol type="1">
    <li>BSCIT</li>
    <li>MSCIT
      <ol type="a">
        <li>Semester 1
          <ul type="square">
            <li>Subject1</li>
            <li>Subject2</li>
            <li>Subject3</li>
          </ul>
        </li>
        <li>Semester 2</li>
        <li>Semester 3</li>
      </ol>
    </li>
    <li>BScNT</li>
  </ol>
</div>
```

```
<hr>
<div id="order-list">
  <ol type="A">
    <li>Movies
      <ol type="i">
        <li>Nepali</li>
        <li>Hindi</li>
        <li>English</li>
      </ol>
    </li>
    <li>Music</li>
```

```
<li>Travel</li>
<li>Social Networking</li>
<li>Chatting</li>
</ol>
</div>
</body>
</html>
```

Output:



← → ↻ ⓘ File D:/Arjun%20Mijar(109)%20Lab%20Reports/Web%20Technology/Q16.html

- 1. BSCIT
- 2. MSCIT
 - a. Semester 1
 - Subject1
 - Subject2
 - Subject3
 - b. Semester 2
 - c. Semester 3
- 3. BScNT

- A. Movies
 - i. Nepali
 - ii. Hindi
 - iii. English
- B. Music
- C. Travel
- D. Social Networking
- E. Chatting

17. Design the following single table using HTML

A	B	C	D
			F
E			G
H	J	K	
L	M		

Source code:

```
<!DOCTYPE html>

<html lang="en">

<head>

  <meta charset="UTF-8">

  <meta name="viewport" content="width=device-width, initial-scale=1.0">

  <title>Complex table</title>

</head>

<body>

  <table >

    <tr>

      <td>A</td>

      <td>B</td>

      <td>C</td>

      <td>D</td>

    </tr>

    <tr>

      <td colspan="3" rowspan="2">E</td>

      <td>F</td>

    </tr>

    <tr>

      <td>G</td>

    </tr>

    <tr>

      <td colspan="2">H</td>

      <td>J</td>

    </tr>

    <tr>

      <td>L</td>

      <td>M</td>

    </tr>

  </table>

</body>

</html>
```

```
<td>K</td>
</tr>
<tr>
<td colspan="2">L</td>
<td colspan="2">M</td>
</tr>
</table>
</body>
</html>
```

Output:

← → ↻ ⓘ File D:/Arjun%20Mijar(109)%20Lab%20Reports/Web%20Technology/Q17.html			
A	B	C	D
E			F
			G
H		J	K
L		M	

18. Design the following table using nested tables in HTML.

Logo (image) ABC Company:

Home(link)

About Us(link)

Contact Us(link)

Content 1 News 1 Content 2 News 2

Copyright Facebook, Twitter(link) Powered By

Source code:

```
<!DOCTYPE html>

<html lang="en">

<head>

  <meta charset="UTF-8" />

  <meta name="viewport" content="width=device-width, initial-scale=1.0" />

  <title>Qn.31 Nested Tables</title>

</head>

<body>

<table border="1" width="100%" cellpadding="5" cellspacing="0">

  <tr>

    <th>

    </th>

    <th>ABC Company</th>

  </tr>

  <tr>

    <td>

      <table border="1" width="90%" cellpadding="5" cellspacing="0" align="center" style="margin-top: 10px;">

        <tr>

          <td><a href="#">Home</a></td>

        </tr>

        <tr>

          <td><a href="#">About Us</a></td>
```

```
</tr>

<tr>

  <td><a href="#">Contact Us</a></td>

</tr>

</table>

</td>

<td>

  <table border="1" width="90%" cellpadding="5" cellspacing="0">

    <tr>

      <td>Content 1</td>

      <td>News 1</td>

    </tr>

    <tr>

      <td>Content 2</td>

      <td>News 2</td>

    </tr>

  </table>

</td>

</tr>

<tr>

  <td>Copyright</td>

  <td>

    <table border="1" width="100%" cellpadding="5" cellspacing="0">

      <tr>

        <td>

          <a href="#">Facebook</a>, <a href="#">Twitter</a>

        </td>

        <td>

          Powered By

        </td>

      </tr>

    </table>

  </td>

</tr>
```

```
</tr>
</table>
</td>
</tr>
</table>
</body>
</html>
```

Output:

		ABC Company	
Home		Content 1	News 1
About Us		Content 2	News 2
Contact Us			
Copyright		Facebook , Twitter	Powered By

19. Design an html form to demonstrate the 10 different types of generated using INPUT tag.

Source code:

```
<!DOCTYPE html>

<html>

<head>

  <meta charset="UTF-8">

  <meta name="viewport" content="width=device-width, initial-scale=1.0">

  <title>Qn.32 10 Different types of controls generated by Input tag</title>

</head>

<body>

  <h2>HTML Form with Different Input Types</h2>

  <form action="#" method="POST">

    <!-- Text Input -->

    <label for="text-input">Text Input:</label>

    <input type="text" id="text-input" name="text-input" placeholder="Enter text"><br><br>

    <!-- Password Input -->

    <label for="password-input">Password Input:</label>

    <input type="password" id="password-input" name="password-input" placeholder="Enter password"><br><br>

    <!-- Radio Buttons -->

    <label>Radio Buttons:</label><br>

    <input type="radio" id="radio1" name="radio-group" value="option1">

    <label for="radio1">Option 1</label>

    <input type="radio" id="radio2" name="radio-group" value="option2">

    <label for="radio2">Option 2</label><br><br>

    <!-- Checkboxes -->
```

```
<label>Checkboxes:</label><br>
<input type="checkbox" id="checkbox1" name="checkbox1" value="checkbox1">
<label for="checkbox1">Checkbox 1</label>
<input type="checkbox" id="checkbox2" name="checkbox2" value="checkbox2">
<label for="checkbox2">Checkbox 2</label><br><br>

<!-- Number Input -->
<label for="number-input">Number Input:</label>
<input type="number" id="number-input" name="number-input" min="1" max="10"><br><br>

<!-- Date Input -->
<label for="date-input">Date Input:</label>
<input type="date" id="date-input" name="date-input"><br><br>

<!-- Color Input -->
<label for="color-input">Color Input:</label>
<input type="color" id="color-input" name="color-input"><br><br>

<!-- File Input -->
<label for="file-input">File Input:</label>
<input type="file" id="file-input" name="file-input"><br><br>

<!-- Range Input -->
<label for="range-input">Range Input:</label>
<input type="range" id="range-input" name="range-input" min="0" max="100"><br><br>

<!-- Email Input -->
<label for="email-input">Email Input:</label>
<input type="email" id="email-input" name="email-input" placeholder="Enter email"><br><br>
```

```
<!-- Submit Button -->

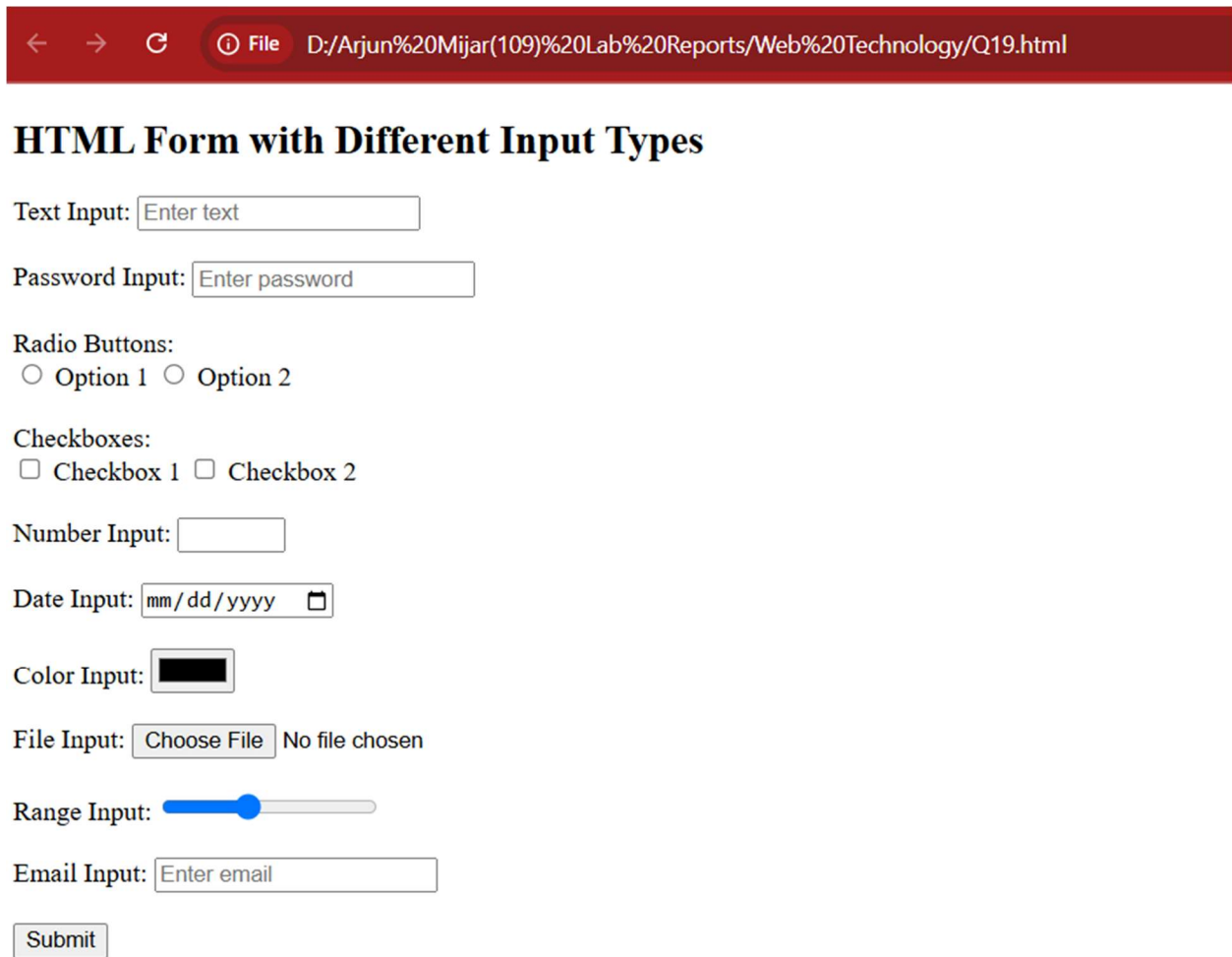
<input type="submit" value="Submit">

</form>

</body>

</html>
```

Output:



The screenshot shows a web browser window with a dark red address bar. The address bar contains a back arrow, a forward arrow, a refresh icon, and a file path: `D:/Arjun%20Mijar(109)%20Lab%20Reports/Web%20Technology/Q19.html`. The main content area has a white background and a title **HTML Form with Different Input Types**. Below the title, there are several form elements: a text input field with placeholder text 'Enter text'; a password input field with placeholder text 'Enter password'; two radio buttons labeled 'Option 1' and 'Option 2'; two unchecked checkboxes labeled 'Checkbox 1' and 'Checkbox 2'; a number input field; a date input field with placeholder text 'mm/dd/yyyy' and a calendar icon; a color input field with a black color swatch; a file input field with a 'Choose File' button and the text 'No file chosen'; a range input field with a blue slider; an email input field with placeholder text 'Enter email'; and a 'Submit' button at the bottom.

Text Input:

Password Input:

Radio Buttons:
☐ Option 1 ☐ Option 2

Checkboxes:
☐ Checkbox 1 ☐ Checkbox 2

Number Input:

Date Input: 

Color Input:

File Input: No file chosen

Range Input:

Email Input:

20. Design an html form to submit a job application form. Demonstrate use of drop-down list, text areas and file upload control.

Source code:

```
<!DOCTYPE html>

<html>

  <head>

    <meta charset="UTF-8" />

    <meta name="viewport" content="width=device-width, initial-scale=1.0" />

    <title>Qn.33 Submit a job application form</title>

  </head>

  <body>

    <h1>Job Application Form</h1>

    <form action="#" method="post" enctype="multipart/form-data">

      <p>

        <label for="name">Full Name:</label>

        <input type="text" id="name" name="name" placeholder="Enter your full name" required/>

      </p>

      <p>

        <label for="email">Email Address:</label>

        <input type="email" id="email" name="email" placeholder="Enter your valid email address"
required/>

      </p>

      <p>

        <label for="phone">Phone Number:</label>

        <input type="tel" id="phone" name="phone" placeholder="Enter your phone number"/>

      </p>

      <p>

        <label for="position">Job Position:</label>

        <select name="position" id="position" required>
```

<option value="">-- Please select a position --</option>
<option value="developer">Software Developer</option>
<option value="designer">Graphic Designer</option>
<option value="writer">Content Writer</option>
<option value="other">Other</option>
</select>

</p>

<p>

<label for="experience">Years of Experience:</label>

<input type="number" id="experience" name="experience" min="0" placeholder="Enter your years of experience"/>

</p>

<p>

<label for="skills">Skills and Technologies:</label>

<textarea name="skills" id="skills" rows="5" placeholder="List your relevant skills and technologies"></textarea>

</p>

<p>

<label for="resume">Upload Resume (PDF or DOCX):</label>

<input type="file" id="resume" name="resume" accept="application/pdf,.docx" required/>

</p>

<p>

<label for="cover_letter">Cover Letter (Optional):</label>

<textarea name="cover_letter" id="cover_letter" rows="5" placeholder="Write a brief cover letter (optional)"></textarea>

</p>

```
<button type="submit">Submit Application</button>

</form>

</body>

</html>
```

Output:

 **File** D:/Arjun%20Mijar(109)%20Lab%20Reports/Web%20Technology/20.html

Job Application Form

Full Name:

Email Address:

Phone Number:

Job Position:

Years of Experience:

Skills and Technologies:

List your relevant skills and technologies

Upload Resume (PDF or DOCX): No file chosen

Cover Letter (Optional):

Write a brief cover letter (optional)

CSS

1. Illustrate the use of background properties in CSS (for image) with four paragraph and style using CSS.

Source code:

```
<!DOCTYPE html>

<html>

<head>

  <meta charset="UTF-8" />

  <meta name="viewport" content="width=device-width, initial-scale=1.0" />

  <title>Q.1 Background Image</title>

  <style>

    body {

      background-image: url("images/2.jpg");

      background-size: cover;

      background-repeat: no-repeat;

      background-position: center center;

    }

    p {

      color: #290341;

      font-size: 20px;

      line-height: 1.5;

      margin-bottom: 20px;

      font-weight: 900;

    }

  </style>

</head>

<body>

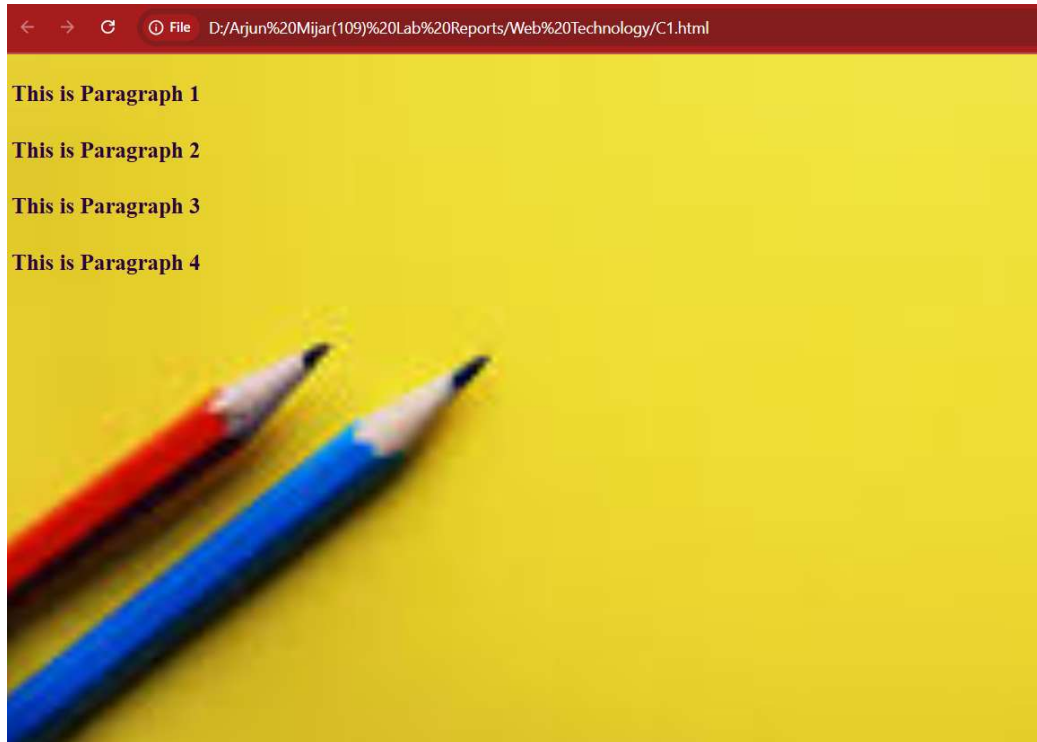
  <p>This is Paragraph 1</p>

  <p>This is Paragraph 2</p>

  <p>This is Paragraph 3</p>
```

```
<p>This is Paragraph 4</p>  
</body>  
</html>
```

Output:



2. Design a web page with six images and style them using external CSS

Source code:

```
<!DOCTYPE html>

<html lang="en">

<head>

  <meta charset="UTF-8">

  <meta name="viewport" content="width=device-width, initial-scale=1.0">

  <title>Images</title>

  <link rel="stylesheet" href="style.css">

</head>

<body>

  <h3>Images:</h3>

  <div class="img-container">

  </div>

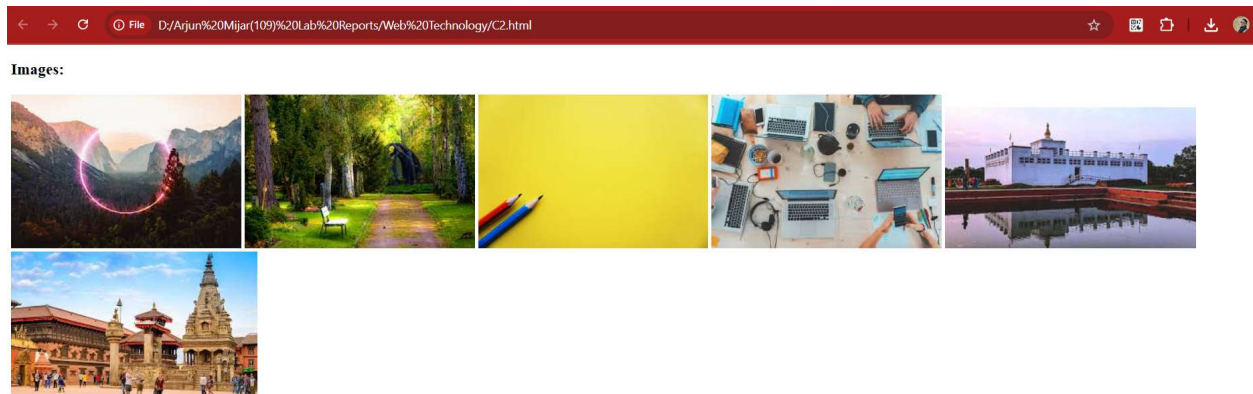
</body>

</html>
```

```
.img-container {
  display: flex;
  justify-content: space-between;
  align-items: center;
}
```

```
.image-container img {  
  width: 150px;  
  height: 100px;  
  border: 2px solid #555;  
  margin: 5px;  
  object-fit: cover;  
}
```

Output:



3. Make a nested list using internal CSS

Source code:

```
<!DOCTYPE html>

<html>

  <head>

    <meta charset="UTF-8" />

    <meta name="viewport" content="width=device-width, initial-scale=1.0" />

    <title>Q.4 Nested List</title>

    <style>

      ul {

        list-style-type: square;

      }

      ul ul {

        list-style-type: circle;

        margin-left: 20px;

      }

      ul ul ul {

        list-style-type: disc;

        margin-left: 40px;

      }

    </style>

  </head>

  <body>

    <ul>

      <li>Background Concept</li>

      <li>

        BSC. CSIT

      </li>

    </ul>
```

```
<li>Cryptography</li>
<li>
  Web Technology
  <ul>
    <li>HTML</li>
    <li>CSS</li>
  </ul>
</li>
</ul>
</li>
<li>XML and JS</li>
</ul>
</body>
</html>
```

Output:



The screenshot shows a web browser window with a dark red header bar. The address bar displays the file path: `D:/Arjun%20Mijar(109)%20Lab%20Reports/Web%20Technology/C3.html`. Below the header, a list of topics is displayed in a structured format:

- Background Concept
- BSC. CSIT
 - Cryptography
 - Web Technology
 - HTML
 - CSS
- XML and JS

4. Make a web page having 3 rows and 2 columns with all the hyperlinks and style it using external CSS

Source code:

```
<!DOCTYPE html>

<html lang="en">

<head>

  <meta charset="UTF-8">

  <meta name="viewport" content="width=device-width, initial-scale=1.0">

  <title>Links</title>

  <link rel="stylesheet" href="style.css">

</head>

<body>

  <div class="link-grid">

    <a href="#">Link 1</a>

    <a href="#">Link 2</a>

    <a href="#">Link 3</a>

    <a href="#">Link 4</a>

    <a href="#">Link 5</a>

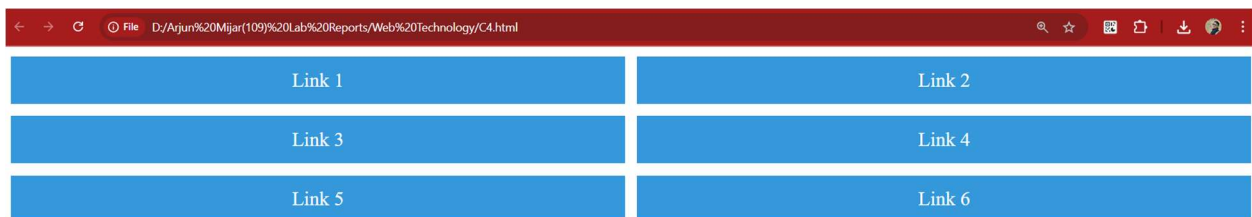
    <a href="#">Link 6</a>

  </div>

</body>

</html>
```

Output:



5. Design an html form to demonstrate the 10 different types of controls generated using INPUT tag. Style the form as:
- o Set the background color of textbox yellow when it is focused.
 - o Change the text color blue to textbox.
 - o Increase the height of combo options when it is checked.
 - o Set the border color red if the input is invalid in textbox textarea.
 - o Set green border to valid input.
 - o Design the form using fieldset

Source code:

```
<!DOCTYPE html>

<html>

<head>

<meta charset="UTF-8" />

<meta name="viewport" content="width=device-width, initial-scale=1.0" />

<title>Q.6 Styled Form</title>

<style>

body {

    font-family: Arial, sans-serif;

}

fieldset {

    border: 2px solid #333;

    padding: 10px;

    margin: 20px;

    width: 300px;

}

label {

    display: block;

    margin-bottom: 5px;

}
```

```
input[type='text']:focus,  
textarea:focus {  
    background-color: yellow;  
    color: blue;  
}
```

```
select:checked {  
    height: 150px;  
}
```

```
input:invalid,  
textarea:invalid {  
    border: 2px solid red;  
}
```

```
input:valid,  
textarea:valid {  
    border: 2px solid green;  
}
```

```
input,  
textarea,  
select {  
    margin-bottom: 10px;  
    width: 100%;  
    padding: 8px;  
    box-sizing: border-box;  
}
```

```
input[type='checkbox'] {
```

```
width: auto;
margin-right: 5px;
}

#radio{
display: flex;
}
</style>
</head>
<body>
<form>
<fieldset>
<legend>Styled Form Controls</legend>

<label for="text">Text:</label>
<input type="text" id="text" required />

<label for="email">Email:</label>
<input type="email" id="email" required />

<label for="password">Password:</label>
<input type="password" id="password" required />

<label for="checkbox">Checkbox:</label>
<input type="checkbox" id="checkbox" />

<label for="radio">Radio:</label>
<div id="radio">
Male<input type="radio" id="radio" />
Female<input type="radio" id="radio" />
```

</div>

<label for="select">Select:</label>

<select id="select" required>

<option value="option1">Option 1</option>

<option value="option2">Option 2</option>

<option value="option3">Option 3</option>

</select>

<label for="textarea">Textarea:</label>

<textarea id="textarea" rows="4" required></textarea>

<label for="date">Date:</label>

<input type="date" id="date" required />

<label for="number">Number:</label>

<input type="number" id="number" required />

<label for="color">Color:</label>

<input type="color" id="color" required />

<input type="submit" value="Submit" />

</fieldset>

</form>

</body>

</html>

Output:



File D:/Arjun%20Mijar(109)%20Lab%20Reports/Web%20Technology/C5.html

Form Controls

Name:

Email:

Password:

Checkbox:

☐

Radio:

Male

☐

Female

☐

Select:

Option 1

Textarea:

Date:

mm / dd / yyyy

Number:

Color:

Submit

6. Design the given table and style as:

Class A Class B

Class C Class D

o Set the border of any style and color.

o Set width of all Classes 600 pixel and height 550 pixel.

o Set different background color for all classes.

o Every text color should be different.

o Display Class C and Class D in uppercase and remaining in lower case.

Source code:

```
<!DOCTYPE html>
```

```
<html>
```

```
<head>
```

```
<meta charset="UTF-8" />
```

```
<meta name="viewport" content="width=device-width, initial-scale=1.0" />
```

```
<title>Styled Table</title>
```

```
<style>
```

```
table {
```

```
border: 2px solid #333;
```

```
width: 600px;
```

```
height: 550px;
```

```
border-collapse: collapse;
```

```
}
```

```
th,
```

```
td {
```

```
border: 2px solid #333;
```

```
padding: 10px;
```

```
text-align: center;
```

```
}
```

```
.classA {
```

```
background-color: #ffcc00;
```

```
color: #990000;  
}
```

```
.classB {  
background-color: #66ccff;  
color: #003366;  
}
```

```
.classC {  
background-color: #99ff99;  
color: #006600;  
text-transform: uppercase;  
}
```

```
.classD {  
background-color: #ff9999;  
color: #cc0000;  
text-transform: uppercase;  
}
```

```
</style>
```

```
</head>
```

```
<body>
```

```
<table>
```

```
<tr>
```

```
<th class="classA">Class A</th>
```

```
<th class="classB">Class B</th>
```

```
</tr>
```

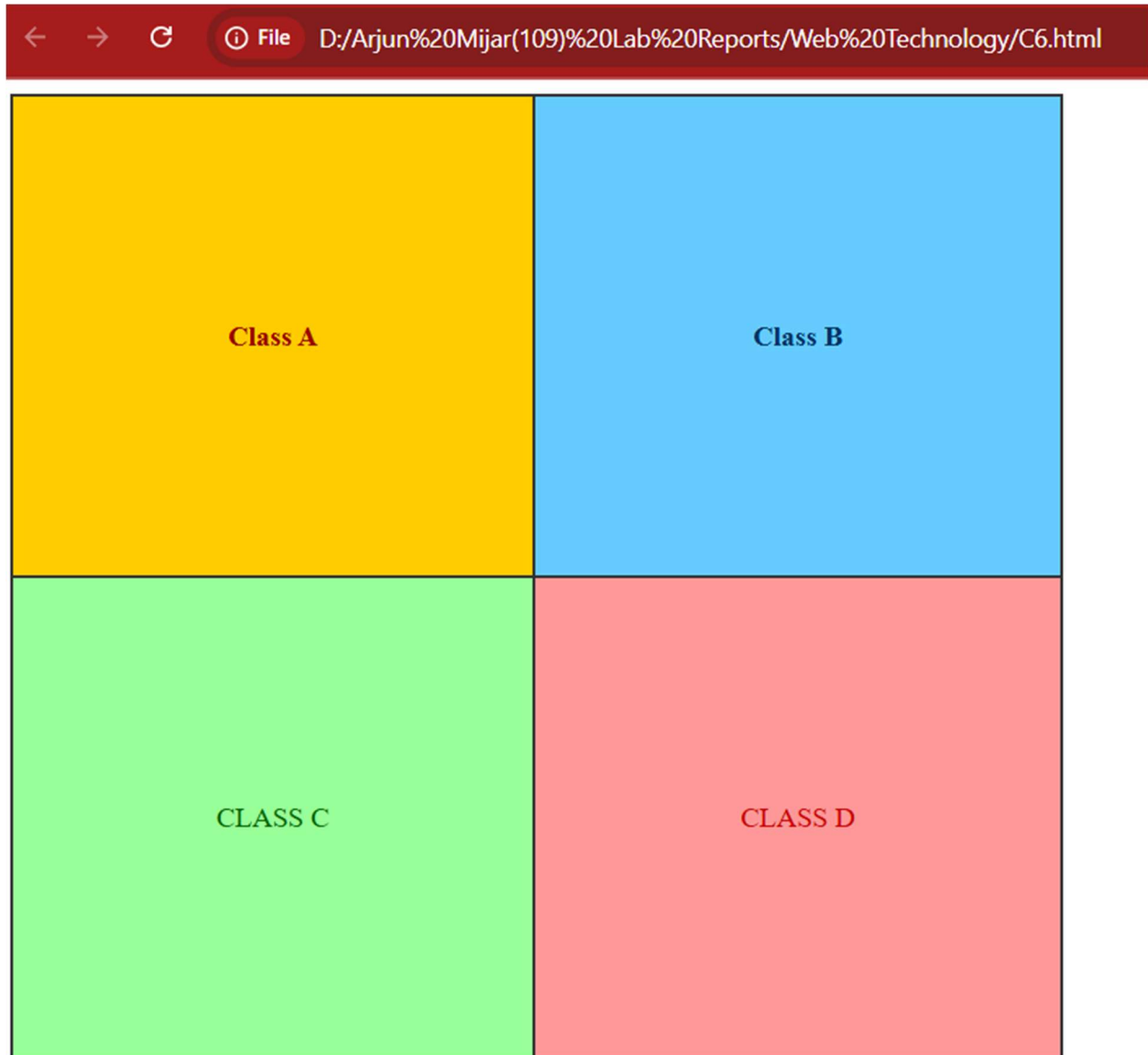
```
<tr>
```

```
<td class="classC">Class C</td>
```

```
<td class="classD">Class D</td>
```

```
</tr>
</table>
</body>
</html>
```

Output:



7. Design a horizontal menu of any five options and style as:
- o Set border for all links.
 - o Set different background color and text color.
 - o Change the font color and increase the font size when mouse over it.
 - o Remove the underline from link.
 - o Change all the default behaviour of links.

Source code:

```
<!DOCTYPE html>

<html>

<head>

<meta charset="UTF-8" />

<meta name="viewport" content="width=device-width, initial-scale=1.0" />

<title>Q.8 Horizontal Menu</title>

<style>

ul {

    list-style: none;

    padding: 0;

    margin: 0;

    display: flex;

}

li {

    margin-right: 15px;

}

a {

    text-decoration: none;

    border: 2px solid #333;

    padding: 10px;

    color: #333;

    background-color: #eee;

    font-size: 16px;

}

a:hover {
```

```
background-color: #ccc;
color: #fff;
font-size: 18px;
}
a:active,
a:focus {
outline: none;
}
</style>
</head>
<body>
<p>Horizontal Menu</p>
<ul>
<li><a href="option1.html">Option 1</a></li>
<li><a href="option2.html">Option 2</a></li>
<li><a href="option3.html">Option 3</a></li>
<li><a href="option4.html">Option 4</a></li>
<li><a href="option5.html">Option 5</a></li>
</ul>
</body>
</html>
```

Output:



8. Illustrate the concept of pseudo elements with example. (::after, ::first-letter, ::selection, ::firstline) and Develop a web page with a paragraph and add some contents before the paragraph with red color and yellow background using CSS.

Source code:

```
<!DOCTYPE html>

<html>

<head>

  <meta charset="UTF-8">

  <meta name="viewport" content="width=device-width, initial-scale=1.0">

  <title>Q.9 Styled Paragraph with Contents</title>

  <style>

    body {

      background-color: #fff8dc;

    }

    .highlight {

      background-color: yellow;

      color: red;

      padding: 10px;

    }

    p {

      font-size: 16px;

      line-height: 1.5;

      margin: 20px 0;

    }

    p::first-line {

      color: #3366cc;

      font-weight: bold;

    }

  </style>

</head>

<body>

  <div class="highlight">

    <p>This is a paragraph with some content before the first line of the paragraph.</p>

  </div>

</body>

</html>
```

```
p::first-letter {  
  font-size: 24px;  
  color: #cc3366;  
  float: left;  
  margin-right: 6px;  
}
```

```
p::after {  
  content: ' - Read more';  
  color: #339966;  
  font-style: italic;  
}
```

```
::selection {  
  background-color: #ffff99;  
  color: #333;  
}
```

```
</style>
```

```
</head>
```

```
<body>
```

```
<div class="highlight">
```

```
<p>Content before the paragraph.</p>
```

```
</div>
```

```
<p>This is the main paragraph content.</p>
```

```
<hr>
```

```
<p>
```

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Phasellus ultricies efficitur velit, ut venenatis leo varius vel.

</p>

</body>

</html>

Output:



JAVASCRIPT and XML

1. Illustrate with examples the concept of alert, confirm and popup boxes.

Source code:

```
<!DOCTYPE html>

<html>

<head>

  <title>JavaScript Popups</title>

  <script>

    // Alert Box Example

    function showAlert() {

      alert("This is an Alert box!\nIt gives information to the user.");

    }


    // Confirm Box Example

    function showConfirm() {

      let result = confirm("Do you want to continue?");

      if (result) {

        alert("You pressed OK!");

      } else {

        alert("You pressed Cancel!");

      }

    }


    // Prompt Box Example

    function showPrompt() {

      let name = prompt("What is your name?");

      if (name != null && name != "") {

        alert("Hello, " + name + "!");

      } else {

        alert("You didn't enter your name.");

      }

    }

  }

</script>

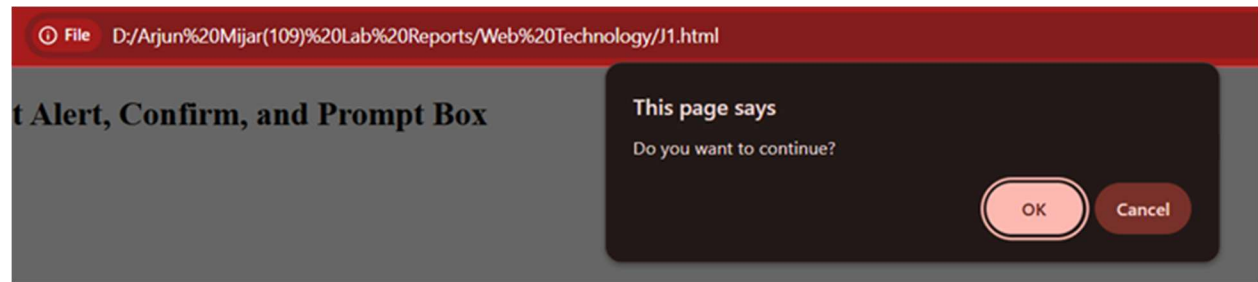
</head>

</html>
```

```
    }  
  }  
</script>  
</head>  
<body>  
  
  <h2>JavaScript Alert, Confirm, and Prompt Box</h2>  
  
  <button onclick="showAlert()">Show Alert</button><br><br>  
  <button onclick="showConfirm()">Show Confirm</button><br><br>  
  <button onclick="showPrompt()">Show Prompt</button>  
  
</body>  
</html>
```

Output:





2. WAP in javascript to demonstrate the objects, its properties and constructors with examples

Source code:

```
<!DOCTYPE html>

<html>

<head>

<title>JavaScript Objects and Constructors</title>

<script>

// 1) Creating a simple object with properties
let student = {
  name: "Arjun",
  age: 22,
  course: "BSc CSIT",
  display: function () {
    return this.name + " is studying " + this.course + " and is " + this.age + " years old.";
  }
};

// 2) Constructor function to create multiple objects
function Car(brand, model, year) {
  this.brand = brand;
  this.model = model;
  this.year = year;

  this.info = function () {
    return "Car: " + this.brand + " " + this.model + " (" + this.year + ")";
  };
}
```

```
// Create objects using the constructor

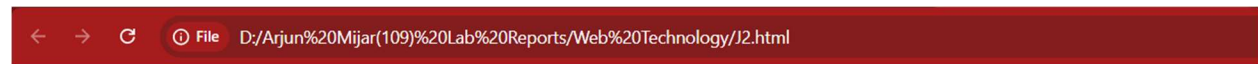
let car1 = new Car("Toyota", "Corolla", 2018);
let car2 = new Car("Hyundai", "i20", 2021);


// Displaying outputs on page
function showObjects() {
    document.getElementById("studentInfo").innerHTML = student.display();
    document.getElementById("car1Info").innerHTML = car1.info();
    document.getElementById("car2Info").innerHTML = car2.info();
}


//console log
console.log(student.display());
console.log(car1.info());
console.log(car2.info());
console.log(student)


</script>
</head>
<body onload="showObjects()">
    <h2>JavaScript Objects, Properties, and Constructors</h2>
    <p id="studentInfo"></p>
    <p id="car1Info"></p>
    <p id="car2Info"></p>
</body>
</html>
```

Output:

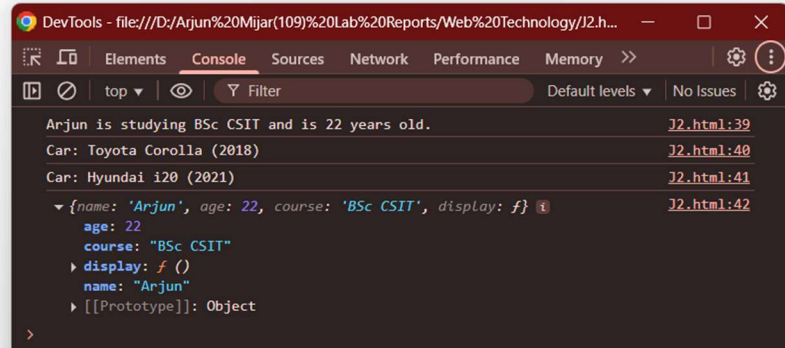


JavaScript Objects, Properties, and Constructors

Arjun is studying BSc CSIT and is 22 years old.

Car: Toyota Corolla (2018)

Car: Hyundai i20 (2021)



3. How array can be created in javascript? Create 2-D array and display the array in HTML table.

Source code:

```
<!DOCTYPE html>

<html>

<head>

  <title>2D Array in HTML Table</title>

  <script>

    // Create a 2D array
    let students = [
      ["Name", "Age", "Grade"],
      ["Ram", 21, "A"],
      ["Shyam", 22, "B+"],
      ["Sita", 20, "A+"]
    ];

    // Function to display 2D array in a table
    function displayTable() {
      let tableHTML = "<table border='1' cellpadding='10'>";

      for (let i = 0; i < students.length; i++) {
        tableHTML += "<tr>";
        for (let j = 0; j < students[i].length; j++) {
          if (i === 0) {
            tableHTML += "<th>" + students[i][j] + "</th>"; // Header
          } else {
            tableHTML += "<td>" + students[i][j] + "</td>"; // Data
          }
        }
      }
      tableHTML += "</tr>";
```

```
}

tableHTML += "</table>";
document.getElementById("output").innerHTML = tableHTML;
}
</script>
</head>
<body onload="displayTable()">

<h2>Displaying 2D Array as Table</h2>
<div id="output"></div>

</body>
</html>
```

Output:



Displaying 2D Array as Table

Name	Age	Grade
Ram	21	A
Shyam	22	B+
Sita	20	A+

4. Explain all the following built in objects with examples of each:

- o Array Methods**
- o Date Methods**
- o String Methods**
- o Math Methods**

Source code:

```
<!DOCTYPE html>

<html>

  <head>

    <meta charset="UTF-8" />

    <meta name="viewport" content="width=device-width, initial-scale=1.0" />

    <title>Builtin Objects</title>

  </head>

  <body>

    <script>

      var colors = ['red', 'green', 'blue'];

      // Array methods

      colors.push('yellow'); // Add element to the end

      colors.pop(); // Remove element from the end

      colors.unshift('orange'); // Add element to the beginning

      colors.shift(); // Remove element from the beginning

      document.write('<h1>Array Methods</h1>');

      document.write('<p>colors = ' + colors + '</p>');

      var currentDate = new Date();

      document.write('<h1>Date Methods</h1>');

      document.write('<p>currentDate = ' + currentDate + '</br>');

      document.write('currentDate.getFullYear() = ' + currentDate.getFullYear() + '</br>');

      document.write('currentDate.getMonth() = ' + currentDate.getMonth() + '</br>');

      document.write('currentDate.getDate() = ' + currentDate.getDate() + '</p>');

      var message = 'Hello, World!';

      document.write('<h1>String Methods</h1>');
```

```
document.write('<p>message = ' + message + '</br>');  
document.write('message.length = ' + message.length + '</br>');  
document.write('message.toUpperCase() = ' + message.toUpperCase() + '</br>');  
document.write('message.indexOf("World") = ' + message.indexOf('World') + '</p>');  
  
document.write('<h1>Math Methods</h1>');  
document.write('<p>Math.random() = ' + Math.random() + '</br>');  
document.write('Math.round(4.7) = ' + Math.round(4.7) + '</br>');  
document.write('Math.pow(2, 3) = ' + Math.pow(2, 3) + '</p>');  
  
</script>  
</body>  
</html>
```

Output:



Array Methods

```
colors = red,green,blue
```

Date Methods

```
currentDate = Mon Jun 23 2025 06:22:07 GMT+0545 (Nepal Time)  
currentDate.getFullYear() = 2025  
currentDate.getMonth() = 5  
currentDate.getDate() = 23
```

String Methods

```
message = Hello, World!  
message.length = 13  
message.toUpperCase() = HELLO, WORLD!  
message.indexOf("World") = 7
```

Math Methods

```
Math.random() = 0.04230531965523421  
Math.round(4.7) = 5  
Math.pow(2, 3) = 8
```

5. WAP that contains a text field for name, age, phone number, email, dropdown for colleges, radio for gender and a checkboxes for i accept all the terms. Perform the client side validations as:
- All fields are required
 - Name must contains alphabets and white spaces and of length greater than 6 characters long
 - Age must be integer and must be between and 50
 - Phone number must be exactly 10 digits long and must start with 98 or 97
 - Email must be valid

Source code:

```
<!DOCTYPE html>

<html lang="en">

<head>

<meta charset="UTF-8" />

<meta name="viewport" content="width=device-width, initial-scale=1.0" />

<title>Form Validation</title>

<style>

.error {

    color: red;

}

</style>

</head>

<body>

<h2>Form Validation</h2>

<form id="myForm" onsubmit="validateForm()">

    <label for="name">Name:</label>

    <input type="text" id="name" name="name" />

    <span id="nameError" class="error"></span><br />

    <label for="age">Age:</label>

    <input type="text" id="age" name="age" />

    <span id="ageError" class="error"></span><br />
```

```
<label for="phone">Phone Number:</label>
<input type="text" id="phone" name="phone" />
<span id="phoneError" class="error"></span><br />
```

```
<label for="email">Email:</label>
<input type="text" id="email" name="email" />
<span id="emailError" class="error"></span><br />
```

```
<label for="colleges">Colleges:</label>
<select id="colleges" name="colleges">
  <option value="" disabled selected>Select a college</option>
  <option value="college1">Amrit Campus</option>
  <option value="college2">Ambition College</option> </select>
<br />
```

```
<label>Gender:</label>
<input type="radio" id="male" name="gender" value="male" /> Male
<input type="radio" id="female" name="gender" value="female" /> Female
<input type="radio" id="other" name="gender" value="other" /> Other<br />
<span id="genderError" class="error"></span><br />
```

```
<label>
  <input type="checkbox" id="terms" name="terms" /> I accept all the terms
</label>
<span id="termsError" class="error"></span><br />
```

```
<input type="submit" value="Submit" />
</form>
```

```
<script>
```

```
function validateForm() {  
  
    var name = document.getElementById('name').value;  
    var age = document.getElementById('age').value;  
    var phone = document.getElementById('phone').value;  
    var email = document.getElementById('email').value;  
    var colleges = document.getElementById('colleges').value;  
    var gender = document.querySelector('input[name="gender"]:checked');  
    var terms = document.getElementById('terms').checked;  
  
    if (name.length < 7 || !/^[a-zA-Z\s]+$/.test(name)) {  
        alert('Name must be at least 7 characters long and contain only alphabets and spaces.');        return false;  
    }  
  
    if (!/^\d+$/.test(age) || age < 18 || age > 50) {  
        alert('Age must be an integer between 18 and 50.');        return false;  
    }  
  
    if (!/^[9][78]\d{8}$/.test(phone)) {  
        alert('Phone number must be exactly 10 digits long and start with 98 or 97.');        return false;  
    }  
  
    if (!/^[S+@S+\.S+$/.test(email)) {  
        alert('Invalid email address.');        return false;  
    }  
  
    if (colleges === "") {
```

```
        alert('Please select a college.');
```

```
        return false;
```

```
    }
```



```
    if (!gender) {
```

```
        alert('Please select a gender.');
```

```
        return false;
```

```
    }
```



```
    if (!terms) {
```

```
        alert('Please accept the terms.');
```

```
        return false;
```

```
    }
```



```
    alert('Form submitted successfully!');
```

```
    return true;
```

```
    }
```

```
</script>
```

```
</body>
```

```
</html>
```

Output:

Name

← → ↻ ⓘ File D:/Arjun%20Mijar(109)%20Lab%20Reports/Web%20Technology/J5.html?name=&age=&phone=&email=

Form Validation

Name:

Age:

Phone Number:

Email:

Colleges:

Gender: ☒ Male ☐ Female ☐ Other

☒ I accept all the terms

This page says

Name must be at least 7 characters long and contain only alphabets and spaces.

Number

← → ↻ ⓘ File D:/Arjun%20Mijar(109)%20Lab%20Reports/Web%20Technology/J5.html?name=Arjun&age=24&phone=9878676789&email=arjun%4

Form Validation

Name:

Age:

Phone Number:

Email:

Colleges:

Gender: ☒ Male ☐ Female ☐ Other

This page says
Phone number must be exactly 10 digits long and start with 98 or 97.

Email

← → ↻ ⓘ File D:/Arjun%20Mijar(109)%20Lab%20Reports/Web%20Technology/J5.html?name=Arjun+Mijar&age=24&phone=9878676789&email=arjun%4

Form Validation

Name:

Age:

Phone Number:

Email:

Colleges:

Gender: ☒ Male ☐ Female ☐ Other

☒ I accept all the terms

This page says
Invalid email address.

Gender

← → ↻ ⓘ File D:/Arjun%20Mijar(109)%20Lab%20Reports/Web%20Technology/J5.html?name=Arjun+Mijar&age=24&phone=9878676789&email=arjungmail.

Form Validation

Name:

Age:

Phone Number:

Email:

Colleges:

Gender: ☐ Male ☐ Female ☐ Other

☒ I accept all the terms

This page says
Please select a gender.

← → ↻ ⓘ File D:/Arjun%20Mijar(109)%20Lab%20Reports/Web%20Technology/J5.html

Form Validation

Name:

Age:

Phone Number:

Email:

Colleges:

Gender: ☒ Male ☐ Female ☐ Other

☒ I accept all the terms

This page says
Form submitted successfully!

6. WAP to show the concept of Try/Catch

Source code:

```
<!DOCTYPE html>

<html>

  <head>

    <meta charset="UTF-8" />

    <meta name="viewport" content="width=device-width, initial-scale=1.0" />

    <title>Try/Catch Example</title>

  </head>

  <body>

    <script>

      try {

        var x = y + 5;

        document.write('<p>x=' + x + 'y=' + y + '</p>');

        document.write('<p>This line will not be executed due to the error.</p>');

      } catch (error) {

        document.write('<p>An error occurred: ' + error + '</p>');

        if (error instanceof ReferenceError) {

          document.write('<p>ReferenceError occurred. Handle it accordingly.</p>');

        }

      } finally {

        document.write('This line will be executed no matter what.</p>');

      }

    </script>

  </body>

</html>
```

Output:

← → ↻ ⓘ File D:/Arjun%20Mijar(109)%20Lab%20Reports/Web%20Technology/J6.html

An error occurred: ReferenceError: y is not defined

ReferenceError occurred. Handle it accordingly.

This line will be executed no matter what.

7. Write jQuery to show the concept of on click, on mouse out,onkeyup, on form submit events

Source code:

```
<!DOCTYPE html>

<html>

  <head>

    <meta charset="UTF-8" />

    <meta name="viewport" content="width=device-width, initial-scale=1.0" />

    <title>jQuery Event Handling</title>

    <!-- Include jQuery -->

    <script src="https://code.jquery.com/jquery-3.6.4.min.js"></script>

  </head>

  <body>

    <button id="myButton">Click me!</button>

    <p id="myParagraph">Mouse over me!</p>

    <input type="text" id="myInput" placeholder="Type something" />

    <form id="myForm">

      <input type="submit" value="Submit Form" />

    </form>

    <br>

    <script>

      // On Click Event

      $('#myButton').on('click', function () {

        alert('Button clicked!');

      });

    </script>

  </body>

</html>
```

```
// On Mouse out Event
```

```
$('#myParagraph').on('mouseout', function () {  
    $(this).css('background-color', 'red');  
});
```

```
// On Key Up Event
```

```
$('#myInput').on('keyup', function () {  
    var inputValue = $(this).val();  
    alert('Input value:', inputValue);  
});
```

```
// On Form Submit Event
```

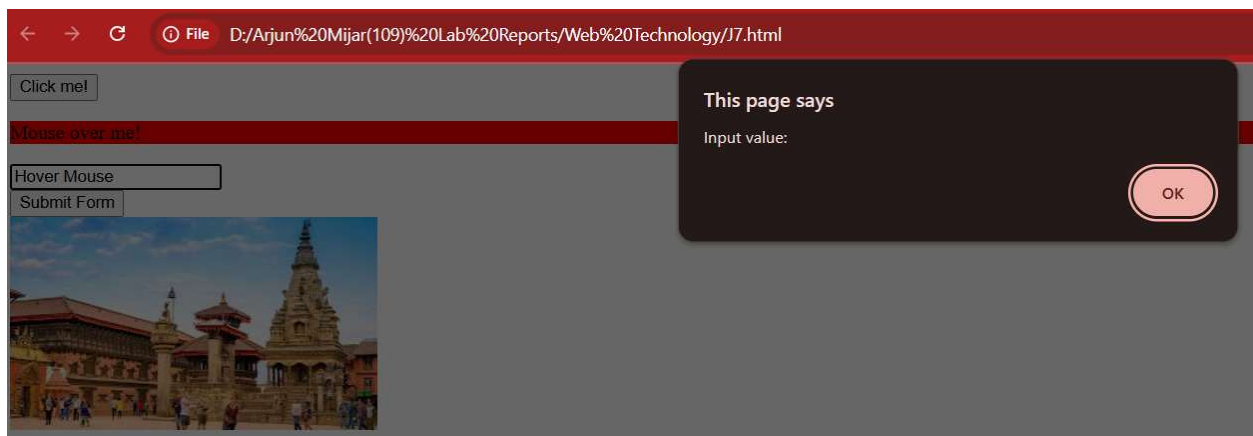
```
$('#myForm').on('submit', function (event) {  
    event.preventDefault();  
    alert('Form submitted!');  
});
```

```
</script>
```

```
</body>
```

```
</html>
```

Output:



8. Illustrate how jQuery effects(hide/show, fade, stop can be done in any image)

Source code:

```
<!DOCTYPE html>

<html>

<head>

  <meta charset="UTF-8" />

  <meta name="viewport" content="width=device-width, initial-scale=1.0" />

  <title>jQuery Effects</title>

  <!-- Include jQuery -->

  <script src="https://code.jquery.com/jquery-3.6.4.min.js"></script>

  <style>

    #myImage {

      width: 200px;

      height: 200px;

      border: 1px solid #ccc;

      margin-bottom: 10px;

    }

  </style>

</head>

<body>

  <button id="showHideButton">Toggle Image</button>

  <button id="fadeInOutButton">Fade In/Out</button>

  <button id="stopButton">Stop Animation</button>


  
```

```

<script>
  // Show/Hide Effect
  $('#showHideButton').on('click', function () {
    $('#myImage').toggle('slow');
  });

  // Fade In/Out Effect
  $('#fadeInOutButton').on('click', function () {
    $('#myImage').fadeToggle('slow');
  });

  // Stop Animation
  $('#stopButton').on('click', function () {
    $('#myImage').stop();
  });
</script>
</body>
</html>

```

Output:



9. How animation is done using jQuery? Illustrate with example

Source code:

```
<!DOCTYPE html>

<html>

  <head>

    <meta charset="UTF-8" />

    <meta name="viewport" content="width=device-width, initial-scale=1.0" />

    <title>jQuery Animation Example</title>

    <script src="https://code.jquery.com/jquery-3.6.4.min.js"></script>

    <style>

      #animatedBox {

        width: 100px;

        height: 100px;

        background-color: blue;

        position: relative;

      }

    </style>

  </head>

  <body>

    <button id="animateButton">Animate Box</button>

    <div id="animatedBox"></div>

    <script>

      $(document).ready(function () {

        // Animate Box when the button is clicked

        $('#animateButton').on('click', function () {

          // Use the animate function to change CSS properties over time

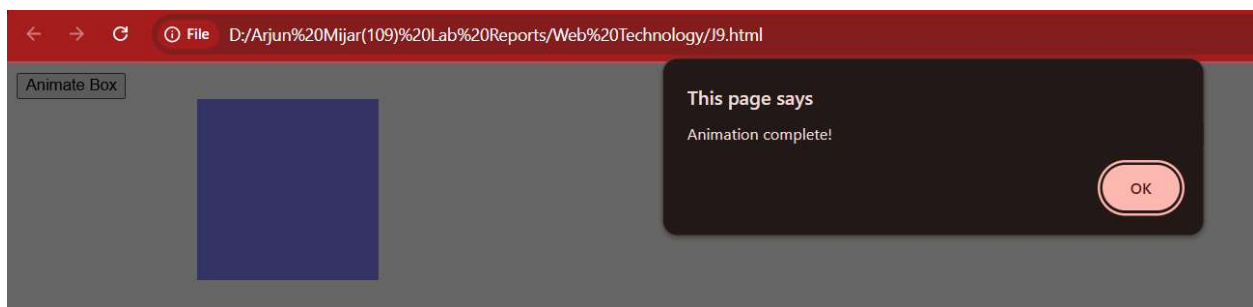
          $('#animatedBox').animate(

            {

              left: '+=150', // Move 150 pixels to the right
```

```
    opacity: 0.5, // Change opacity to 0.5
    width: '+=50', // Increase width by 50 pixels
    height: '+=50', // Increase height by 50 pixels
  },
  1000,
  function () {
    alert('Animation complete!');
  }
);
});
});
</script>
</body>
</html>
```

Output:



10. How json file is created? How can we parse the JSON file using JavaScript?

Source code:

```
<!DOCTYPE html>

<html>

<head>

  <meta charset="UTF-8">

  <meta name="viewport" content="width=device-width, initial-scale=1.0">

  <title>Q.11 Parse JSON with JavaScript</title>

</head>

<body>


<script>

  var jsonString = '{"name": "Suyash Shrestha", "age": 21, "city": "KTM", "skills": ["JavaScript", "HTML", "CSS"]}';

  var parsedData = JSON.parse(jsonString);

  document.write("<p>Name: " + parsedData.name + "</br>");

  document.write("Age: " + parsedData.age + "</br>");

  document.write("City: " + parsedData.city + "</br>");

  document.write("Skills: " + parsedData.skills + "</p>");

</script>

</body>

</html>
```

Output:



11. How AJAX is used to get the data from the server to the client side? Illustrate with example.

Source code:

```
<!DOCTYPE html>

<html>

  <head>

    <meta charset="UTF-8" />

    <meta name="viewport" content="width=device-width, initial-scale=1.0" />

    <title>AJAX Example</title>

  </head>

  <body>

    <button id="getDataButton">Get Data</button>

    <div id="result"></div>


    <script>

      document

      .getElementById('getDataButton')

      .addEventListener('click', function () {

        // Create a new XMLHttpRequest object

        var xhr = new XMLHttpRequest();


        // Configure it to make a GET request to the specified URL

        xhr.open('GET', 'https://jsonplaceholder.typicode.com/todos/5', true);


        // Set up a callback function to handle the response

        xhr.onreadystatechange = function () {

          if (xhr.readyState === 4 && xhr.status === 200) {

            // Parse the JSON response

            var responseData = JSON.parse(xhr.responseText);


            // Update the content on the client side
```

```
document.getElementById('result').innerHTML = `
    <p>User ID: ${responseData.userId}<br/>
    Title: ${responseData.title}<br/>
    Completed: ${responseData.completed}</p>
`;
}
};

// Send the request
xhr.send();
});
</script>
</body>
</html>
```

Output:



12. Create a simple XML element.

Source code:

```
<?xml version="1.0" encoding="UTF-8"?>
<book>
  <title>Introduction to XML</title>
  <author>John Doe</author>
  <publicationYear>2022</publicationYear>
</book>
```

Output:



13. Create an XML document that conforms to the following DTD

Source code:

```
<?xml version="1.0" encoding="UTF-8"?>
<!DOCTYPE Report SYSTEM "report.dtd">
<Report>
  <Students>
    <Student StudId="s1">
      <Name>
        <First>Arjun</First>
        <Last>Mijar</Last>
```

```
</Name>

<Status>Enrolled</Status>

<CrsTaken CrsCode="c1"/>

<CrsTaken CrsCode="c2"/>

</Student>

<Student StudId="s2">

  <Name>

    <First>Aayush</First>

    <Last>Gautam</Last>

  </Name>

  <Status>Graduated</Status>

  <CrsTaken CrsCode="c1"/>

</Student>

</Students>

<Classes>

  <Class>

    <Semester>Fall 2022</Semester>

    <ClassRoster Members="s1 s2"/>

  </Class>

</Classes>

<Courses>

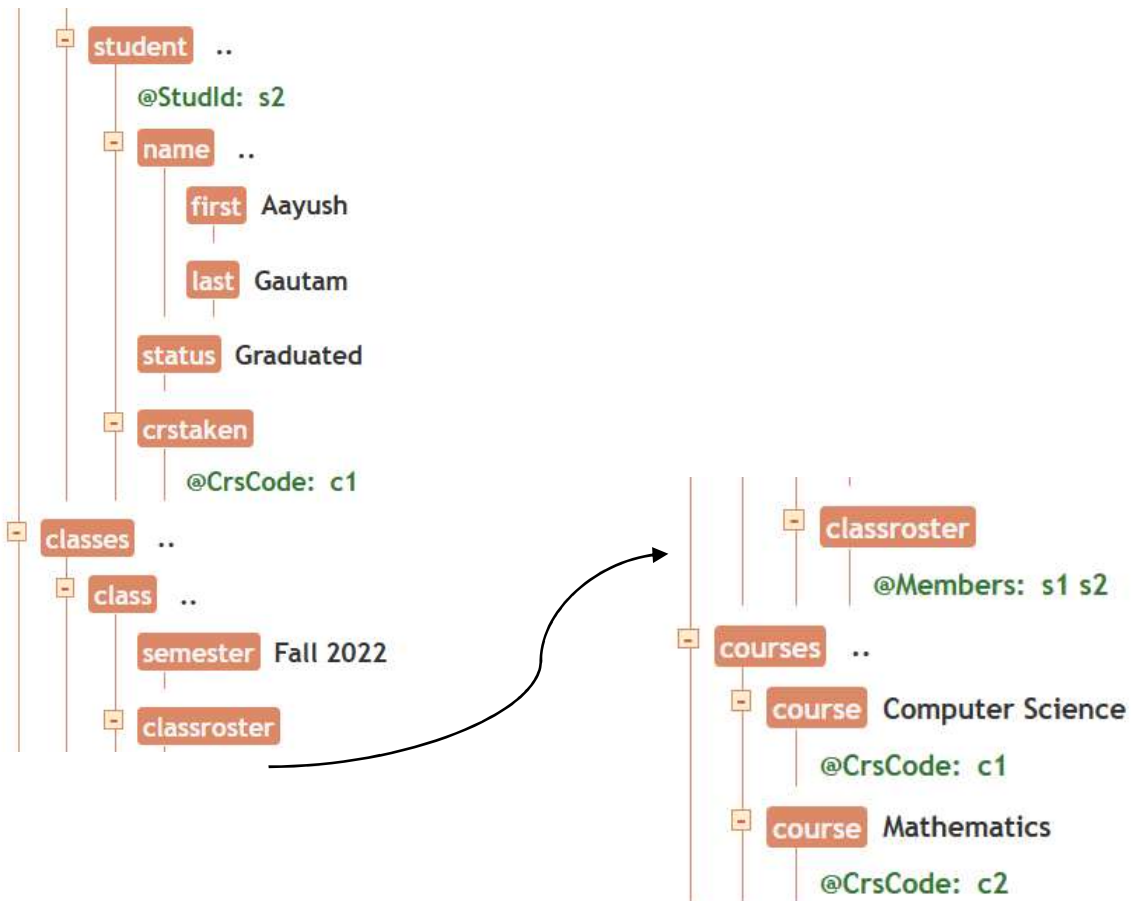
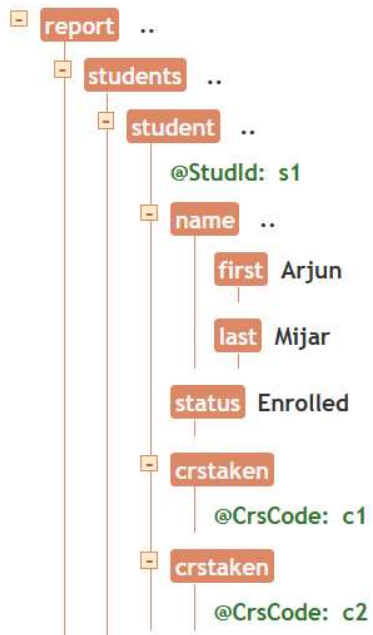
  <Course CrsCode="c1">Computer Science</Course>

  <Course CrsCode="c2">Mathematics</Course>

</Courses>

</Report>
```

Output:



14. Create a XML document with simple and complex type

Source code:

```
<?xml version="1.0" encoding="UTF-8"?>

<library xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xsi:noNamespaceSchemaLocation="library.xsd">

    <!-- Simple Type Example -->

    <book>

        <title>Introduction to XML</title>

        <author>Arjun Mijar</author>

        <publicationYear>2023</publicationYear>

        <price>29.99</price>

    </book>

    <!-- Complex Type Example -->

    <bookstore>

        <book>

            <title>Web Development Basics</title>

            <author>Jane Smith</author>

            <publicationYear>2021</publicationYear>

            <details>

                <genre>Programming</genre>

                <pages>300</pages>

            </details>

        </book>

    </bookstore>

</library>
```

Output:



15. How XSLT is used to transform the XML document? Illustrate with example

Source code:

lab.xml

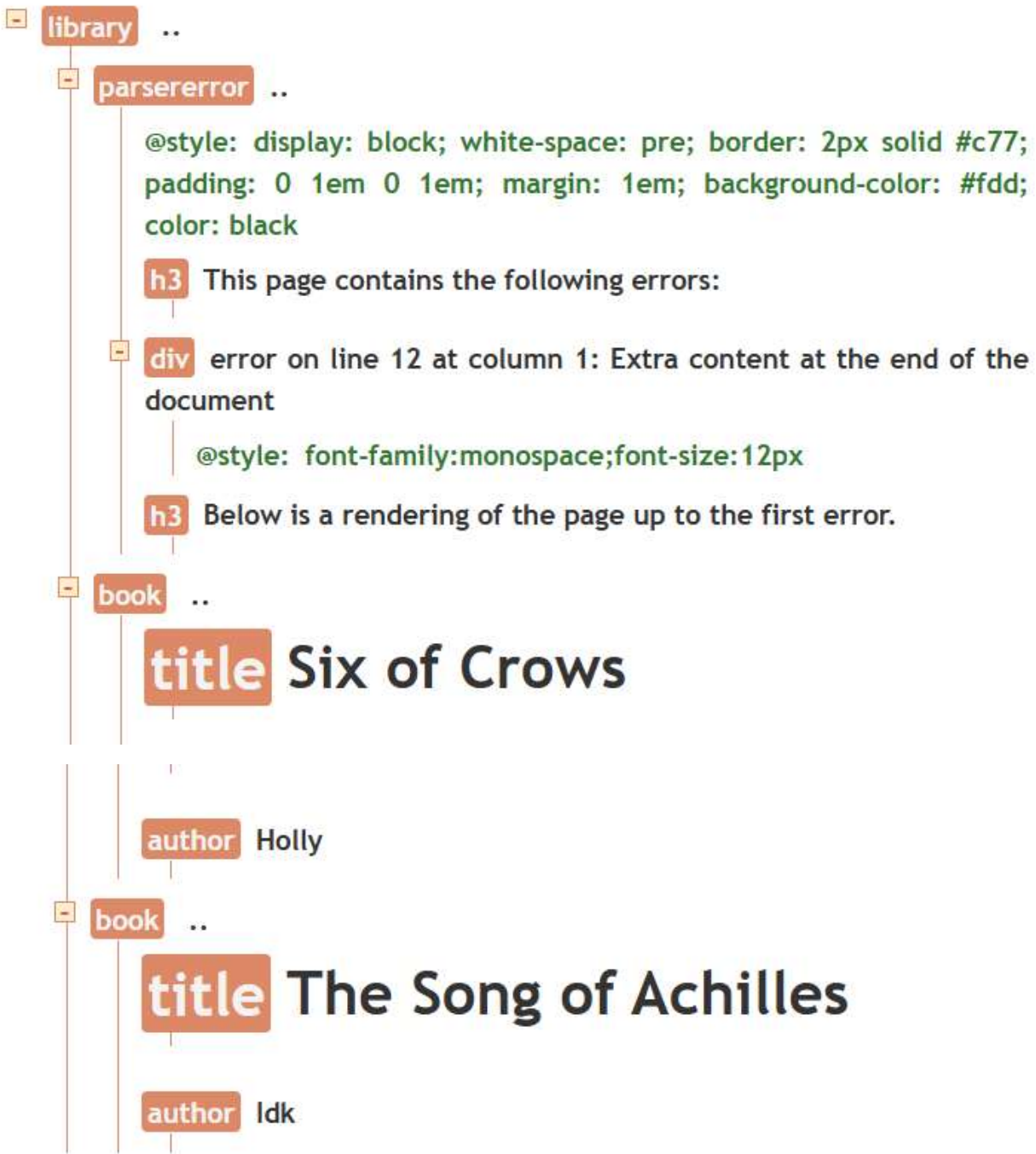
```
<?xml version="1.0" encoding="UTF-8"?> <library> <book> <title>Six of Crows</title>
<author>Holly</author> </book> <book> <title>The Song of Achilles</title>
<author>Idk</author> </book> </library>

lab.html <html> <head> <title>Qn.16</title> </head> <body> <h1>List</h1> <ul> <li>
<strong>Title:</strong> Six if Crows<br /> <strong>Author:</strong> Holly </li> <li>
<strong>Title:</strong> The song of Achilles<br /> <strong>Author:</strong> Idk </li>
</ul> </body></html>
```

transform.xslt

```
<?xml version="1.0" encoding="UTF-8"?> <xsl:stylesheet version="1.0"
xmlns:xsl="http://www.w3.org/1999/XSL/Transform"> <xsl:template match="/"> <html>
<head> <title>List</title> </head> <body> <h1>List</h1> <ul> <xsl:apply-templates
select="library/book"/> </ul> </body> </html> </xsl:template> <xsl:template
match="book"> <li> <strong>Title:</strong> <xsl:value-of select="title"/><br/>
<strong>Author:</strong> <xsl:value-of select="author"/> </li>
</xsl:template></xsl:stylesheet>
```

Output:



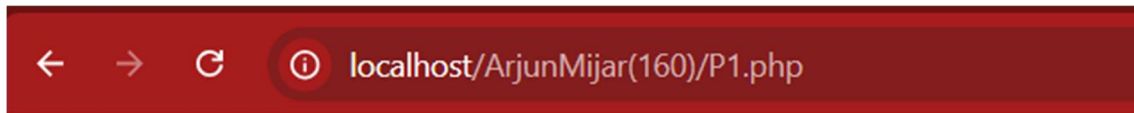
PHP

1. Declare variables of different data types (integer, float, string) and demonstrate the usage of constants in PHP .

Source code:

```
<?php
$integerVal=160;
$floatVal=1.60;
$stringVar="Hello, PHP!: I am Arjun Mijar.";
define("My_Constant","This is constant");
echo My_Constant . "<br>";
echo $integerVal . "<br>";
echo $floatVal . "<br>";
echo $stringVar;
?>
```

Output:



This is constant
160
1.6
Hello, PHP!: I am Arjun Mijar.

2. Create a PHP program that uses if-else statements to check if a number is positive, negative, or zero. Utilize different operators for comparison.

Source code:

```
<?php
$num=160;

if($num>0){
    echo "Positive Number.";
}
else if($num=0)
    echo "Zero";

else
    echo "Negative Number.";
?>
```

Output:



Positive Number.

3. Develop a function that calculates the average of an array of numbers. Test the function with various arrays

Source code:

```
<?php
function calculateAverage($numbers)
{
    return array_sum($numbers) / count($numbers);
}
$testArray = [160, 109, 18];
echo calculateAverage($testArray);
?>
```

Output:



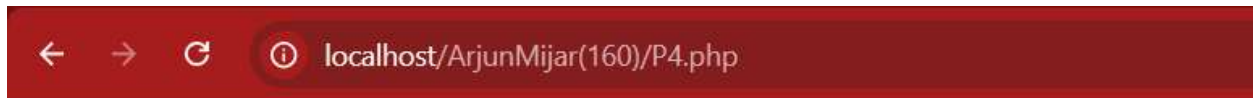
4. Define a PHP class representing a "Book" with properties like title and author. Create objects of this class and display their attributes.

Soruce code:

```
<?php
class Book
{
    public $title;
    public $author;
}

$book1 = new Book();
$book1->title = "The Poor Dad Rich Dad";
$book1->author = "Robert Kiyosaki";
echo $book1->title . " by " . $book1->author;
?>
```

Output:



The Poor Dad Rich Dad by Robert Kiyosaki

5. Design a simple HTML form with input fields. Create a PHP script that retrieves and displays the form data when submitted.

Source code:

```
<html>

<head>

    <title>Form retrieval</title>

</head>


<body>

    <form method="post" action="#">

        <input type="text" name="username" placeholder="Username">

        <input type="submit" value="Submit">

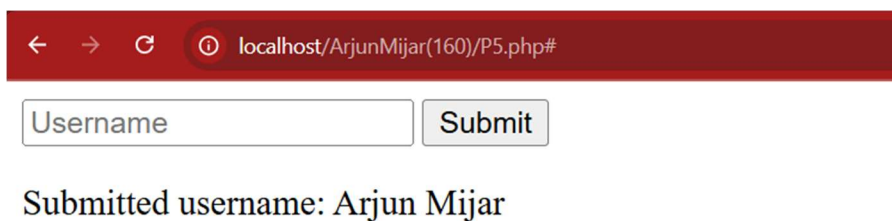
    </form>

</body>

</html>


<?php
if ($_SERVER["REQUEST_METHOD"] == "POST") {
    $username = $_POST["username"];
    echo "Submitted username: $username";
}
?>
```

Output:

A screenshot of a web browser window. The address bar shows the URL 'localhost/ArjunMijar(160)/P5.php#'. Below the address bar, there is a text input field with the placeholder text 'Username' and a 'Submit' button. Below the form, the text 'Submitted username: Arjun Mijar' is displayed.

← → ↻ ⓘ localhost/ArjunMijar(160)/P5.php#

Username Submit

Submitted username: Arjun Mijar

- 6. Enhance the previous form with PHP validation. Ensure that required fields are filled, and validate email addresses and make sure the phone number starts with 98 or 97 using regex**

Source code:

```
<!DOCTYPE html>

<html>

<head>

  <title>PHP Form Validation</title>

  <style>

    body { font-family: Arial; background-color: #f9f9f9; padding: 30px; }

    form { background: #fff; padding: 20px; border-radius: 10px; width: 400px; margin: auto; box-shadow:
    0 0 10px #ccc; }

    input, select { width: 100%; padding: 10px; margin: 10px 0; }

    .error { color: red; font-size: 14px; }

  </style>

</head>

<body>

<?php
// Define variables and initialize
$name = $email = $phone = "";
$nameErr = $emailErr = $phoneErr = "";
$success = "";

if ($_SERVER["REQUEST_METHOD"] == "POST") {

  // Validate name
  if (empty($_POST["name"])) {
    $nameErr = "Name is required";
  } else {
    $name = trim($_POST["name"]);
    if (!preg_match("/^[a-zA-Z\s]{6,}$/", $name)) {
```

```

        $nameErr = "Name must be at least 6 characters and contain only letters and spaces";
    }
}

// Validate email
if (empty($_POST["email"])) {
    $emailErr = "Email is required";
} else {
    $email = trim($_POST["email"]);
    if (!filter_var($email, FILTER_VALIDATE_EMAIL)) {
        $emailErr = "Invalid email format";
    }
}

// Validate phone
if (empty($_POST["phone"])) {
    $phoneErr = "Phone number is required";
} else {
    $phone = trim($_POST["phone"]);
    if (!preg_match("/^(98|97)\d{8}$/", $phone)) {
        $phoneErr = "Phone must start with 98 or 97 and be 10 digits long";
    }
}

// Final success message
if ($nameErr == "" && $emailErr == "" && $phoneErr == "") {
    $success = "Form submitted successfully!";
}
}
?>

```

```
<h2 align="center">PHP Validated Form</h2>
```

```
<form method="post" action="">
```

```
<label>Name:</label>
```

```
<input type="text" name="name" value="<?= htmlspecialchars($name); ?>">
```

```
<span class="error"><?= $nameErr; ?></span>
```

```
<label>Email:</label>
```

```
<input type="text" name="email" value="<?= htmlspecialchars($email); ?>">
```

```
<span class="error"><?= $emailErr; ?></span>
```

```
<label>Phone:</label>
```

```
<input type="text" name="phone" value="<?= htmlspecialchars($phone); ?>">
```

```
<span class="error"><?= $phoneErr; ?></span>
```

```
<input type="submit" value="Submit">
```

```
</form>
```

```
<?php if ($success): ?>
```

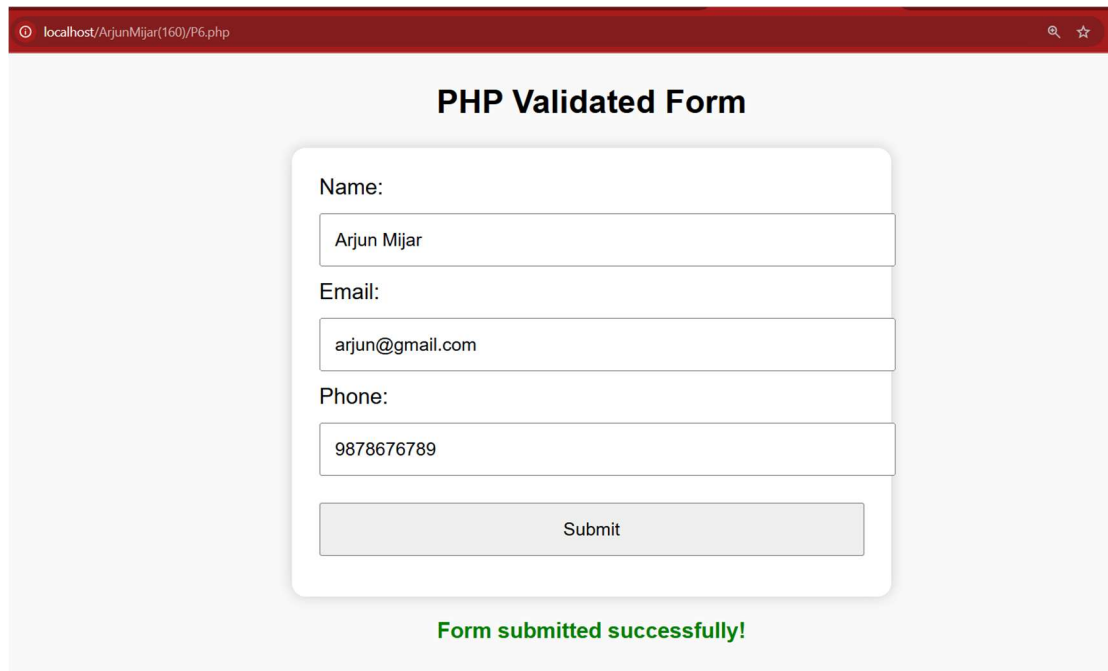
```
<p style="color:green; text-align:center;"><strong><?= $success; ?></strong></p>
```

```
<?php endif; ?>
```

```
</body>
```

```
</html>
```

Output:



The screenshot shows a web browser window with a dark red header bar. The address bar displays "localhost/ArjunMijar(160)/P6.php". The main content area has a light gray background and is titled "PHP Validated Form" in bold black text. Below the title is a white rounded rectangle containing a form. The form has three input fields: "Name:" with the value "Arjun Mijar", "Email:" with the value "arjun@gmail.com", and "Phone:" with the value "9878676789". Below these fields is a gray "Submit" button. At the bottom of the form container, the text "Form submitted successfully!" is displayed in green.

localhost/ArjunMijar(160)/P6.php

PHP Validated Form

Name:
Arjun Mijar

Email:
arjun@gmail.com

Phone:
9878676789

Submit

Form submitted successfully!

7. Implement PHP code that responds to a button click event on a form. Display a message when the button is clicked

Source code:

```
<html>

<head>

    <title>Button click</title>

</head>

<form method="post">

    <input type="submit" name="submitBtn" value="Click me">

</form>

</html>


<?php
if (isset($_POST["submitBtn"])) {
    echo "Button clicked!";
}
?>
```

Output:



Button clicked!

8. Create a PHP script that sets a cookie with user preferences. Retrieve and display this information on subsequent visits

Source code:

```
<?php  
setcookie("user_pref", "dark_mode", time() + 3600, "/");  
  
if (isset($_COOKIE["user_pref"])) {  
    echo "User preference: " . $_COOKIE["user_pref"];  
}  
?>
```

Output:



User preference: dark_mode

9. Build a form with various elements (textboxes, checkboxes, radio buttons). Use PHP to access and display the selected values.

Source code:

```
<!DOCTYPE html>

<html>

<head>

    <title>PHP Form</title>

</head>

<body>

    <h2>PHP Form Example</h2>


    <?php
    // Check if form is submitted
    if ($_SERVER["REQUEST_METHOD"] == "POST") {
        // Retrieve and display the values
        $name = $_POST["name"];
        $email = $_POST["email"];
        $gender = $_POST["gender"];
        $interests = isset($_POST["interests"]) ? $_POST["interests"] : [];

        echo "<h3>Form Submission Results</h3>";
        echo "Name: $name<br/>";
        echo "Email: $email<br/>";
        echo "Gender: $gender<br/>";

        if (!empty($interests)) {
            echo "Interests: " . implode(", ", $interests) . "<br/>";
        } else {
            echo "No interests selected.<br/>";
        }
    }
```

```
}  
?>
```

```
<form action="<?php echo htmlspecialchars($_SERVER["PHP_SELF"]); ?>" method="post">
```

```
  <label for="name">Name:</label>
```

```
  <input type="text" name="name" id="name" required>
```

```
  <br>
```

```
  <label for="email">Email:</label>
```

```
  <input type="email" name="email" id="email" required>
```

```
  <br>
```

```
  <label>Gender:</label>
```

```
  <input type="radio" name="gender" value="male" id="male">
```

```
  <label for="male">Male</label>
```

```
  <input type="radio" name="gender" value="female" id="female">
```

```
  <label for="female">Female</label>
```

```
  <br>
```

```
  <label>Interests:</label>
```

```
  <input type="checkbox" name="interests[]" value="programming" id="programming">
```

```
  <label for="programming">Programming</label>
```

```
  <input type="checkbox" name="interests[]" value="reading" id="reading">
```

```
  <label for="reading">Reading</label>
```

```
  <input type="checkbox" name="interests[]" value="traveling" id="traveling">
```

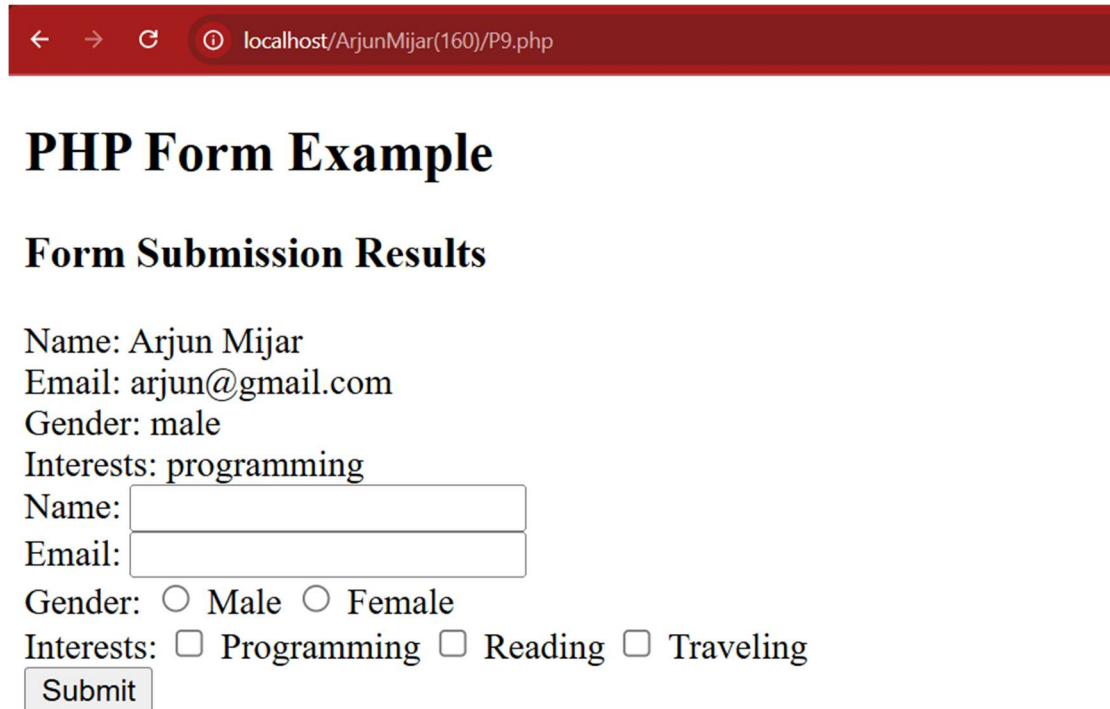
```
  <label for="traveling">Traveling</label>
```

```
  <br>
```

```
  <input type="submit" value="Submit">
```

```
</form>
</body>
</html>
```

Output:



← → ↻ ⓘ localhost/ArjunMijar(160)/P9.php

PHP Form Example

Form Submission Results

Name: Arjun Mijar
Email: arjun@gmail.com
Gender: male
Interests: programming

Name:
Email:

Gender: ☐ Male ☐ Female

Interests: ☐ Programming ☐ Reading ☐ Traveling

10. Develop a PHP script that reads content from a text file, modifies it, and then writes the updated content back to the file

Source code:

```
<?php
$filename = "example.txt";

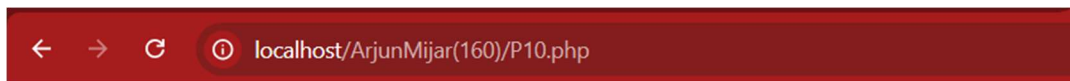
// Read content from file
$content = file_get_contents($filename);

// Modify content
$newContent = str_replace("old_text", "new_text", $content);

// Write updated content back to file
file_put_contents($filename, $newContent);

echo "File updated successfully";
?>
```

Output:



File updated successfully